SemiBuildingChange: A Semi-Supervised High-Resolution Remote Sensing Image Building Change Detection Method With a Pseudo Bitemporal Data Generator

Chengzhe Sun, Hao Chen, Chun Du, and Ning Jing

***Abstract*— Remote sensing (RS) image change detection (CD) aims to obtain change information of the target area between multitemporal RS images. With the modernization of cities, building CD (BCD) plays a pivotal role in land resource planning, smart city construction, and natural disaster assessment, and it is a typical application field of CD task. Recently, deep learning (DL)-based methods have shown their superiority in RS image CD. However, the performance of the existing supervised CD methods relies heavily on a large amount of high-quality annotated bitemporal RS image as training data, which is usually hard to obtain in practice. To address this issue, a semi-supervised BCD method using a pseudo bitemporal data (PITA) generator with consistency regularization was proposed. This method only needs a very small amount of single-temporal RS images with building extraction labels as labeled data. First, with the help of the PITA generator, the model can generate a large number of pseudo bitemporal images with CD labels from a small number of single-temporal images and corresponding building extraction labels automatically, which greatly augments the labeled dataset for CD model training. Then, we proposed an error-prone data enhancement fine-tuning strategy to improve the learning effect of the proposed model to these synthesized training data. Finally, we enhance the robustness of the model by forcing the model to make consistent predictions on the images before and after perturbations. Extensive experimental results demonstrate that our method can effectively improve the BCD performance of the model even if labeled data are scare and outperform the state-of-the-art methods.**

***Index Terms*— Building change detection (BCD), data augmentation, pseudo bitemporal generator, remote sensing (RS), semi-supervised learning (SSL).**

I. Introduction

REMOTE sensing (RS) image change detection (CD) is a meaningful and challenging fundamental task that detects semantic changes in the same area from multitemporal RS images [1]. It usually takes bitemporal RS images as input and pixelwise CD results as output. Nowadays, RS building CD (BCD) has a wide range of applications in the fields of land resource planning [2], smart city construction [3], and natural disaster assessment [4].

With the advent of RS big data and the development of deep learning (DL), numerous RS images' CD methods based on deep neural networks have been proposed, achieving high accurate results. However, most of these methods are supervised, relying on large quantities of high-quality pixelwise labels for effective model training. Unfortunately, manually annotating pixelwise labels for numerous bitemporal RS images is labor-intensive and time-consuming. Therefore, the CD models based on supervised learning usually suffer from either insufficient training data or high costs of manual annotation, limiting their real-world applicability. To address this issue, methods based on unsupervised learning [5], [6], weak supervised learning [7], [8], [9], and semi-supervised learning (SSL) [10], [11], [12] have been proposed.

Unsupervised methods [13] can accomplish CD without any labels, but the achieved accuracy is often insufficient to support practical applications. Weak supervised methods aim to replace accurate pixelwise labels with coarse labels, such as image-level labels [14], bounding boxes [15], or graffiti [16]. While these methods alleviate some of the annotation workload in comparison to supervised methods, the identification of the changes in bitemporal RS images remains a necessary task when generating coarse labels, requiring significant effort, especially for high-resolution RS images. The third kind of methods is SSL, which use a small number of labeled images and a large number of unlabeled images to train CD models. Labeled images can guide the model to understand the task, while unlabeled images enhance the robustness and generalization of the model. Semi-supervised methods can be achieved by autoencoder [17], generative adversarial networks (GANs) [18], and graph neural networks [19] and have been proved to be effective for CD in recent research [20], [21].

Except the above three methods, a single-temporal supervised CD method which is called ChangeStar has been proposed [22]. This method focuses on BCD tasks, using an annotated building extraction dataset to synthesize a bitemporal image's CD dataset for training. Specifically, when training the CD model, the inputs of the CD network are two different single-temporal images instead of bitemporal RS images, while the CD labels are obtained by performing an exclusive-or (XOR) operation on the two corresponding

Manuscript received 14 April 2023; revised 4 August 2023 and 2 September 2023; accepted 25 September 2023. Date of publication 4 October 2023; date of current version 17 October 2023. This work was supported in part by the National Natural Science Foundation of China under Grant U19A2058, Grant 61806211, and Grant 41971362. *(Corresponding author: Hao Chen.)*

The authors are with the College of Electronic Science and Technology, National University of Defense Technology, Changsha, Hunan 430070, China (e-mail: sunchengzhe@nudt.edu.cn; hchen@nudt.edu.cn; duchun@nudt.edu.cn; ningjing@nudt.edu.cn).

Digital Object Identifier 10.1109/TGRS.2023.3321637

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:10:49 UTC from IEEE Xplore. Restrictions apply.

building extraction labels of the two bitemporal RS images. This approach bypasses the manual annotation process of pixelwise change masks from bitemporal images. The study demonstrates the possibility of training an effective BCD model using single-temporal RS images with synthesized CD labels [called pseudo bitemporal data (PITA)]. However, this is still a fully supervised method that requires a large number of building extraction labels, which is also expensive, time-consuming, and labor-intensive to obtain.

Inspired by ChangStar [22], in this article, a novel semi-supervised RS buildings' CD method (called SemiBuildingChange) with a PITA generator is proposed. In comparison to ChangStar [22], first, SemiBuildingChange reduced the number of labels required from thousands to dozens by incorporating a semi-supervised method. Second, SemiBuildingChange proposed a PITA to effectively use these dozens of labels, enabling us to generate a larger number of diverse pseudo bitemporal images. Finally, considering that different pseudo bitemporal images present varying levels of learning difficulty, an error-prone data enhancement fine-tuning strategy was introduced in the proposed PITA generator to enhance our method's performance. The main contributions of this article are as follows.

1) To address the challenge of annotating a large number of bitemporal RS images, we propose a new method called SemiBuildingChange for object CD using single-temporal SSL. To the best of our knowledge, this is the first method that uses single-temporal SSL specifically for BCD. The method achieves high-precision CD while minimizing the cost of labeling. In addition, it can be easily applied to other object CD tasks.
2) A PITA generator with error-prone data enhancement fine-tuning strategy is designed to not only synthesize a large amount of high-quality BCD training data but also prioritize selecting error-prone data for model fine-tuning to further enhance the learning effect of the proposed model.
3) Extensive experimental results on two widely used high-resolution RS datasets demonstrate that our model can effectively improve the accuracy of BCD results and outperform several state-of-the-art methods.

The rest of this article is organized as follows. Section II introduced the related work about CD methods. The proposed semi-supervised method SemiBuildingChange is illustrated in detail in Section III. Experiments and analyses are presented in Section IV. In Section V, we conduct discussions. Section VI draws the conclusions of this article.

II. Related Work

In this section, the traditional CD methods, deep-learning based CD methods, and semi-supervised CD method are briefly illustrated.

A. Traditional CD Methods

Initially, CD was performed through algebraic calculations between image bands [23], such as band difference, band ratio, band regression, and spectral angle. As research progressed, image transformation methods involving feature extraction and data statistics gained popularity for CD. These included PCA interpolation, PCA regression, and joint PCA [24]. However, these approaches primarily focused on pixel-level analysis without incorporating spatial information. Consequently, the resulting CD maps often exhibited misalignment with actual object boundaries and details. The emergence of high-resolution RS images further accentuated these limitations.

To address the richer detail and complex spectral and geometric features present in high-resolution RS images, object-based methods were proposed. These methods divided the image into numerous homogeneous regions, converting the basic processing unit from pixels into segmented blocks. This allowed the utilization of object-level spatial context information, such as texture, shape, and the spatial relationship with neighboring objects. The object-based methods can be further classified into four categories [25]: 1) object-based CD, which compares the segmented objects to detect changes; 2) classification-based CD, which directly compares the categories of segmented objects between two images after classification, relying on classification accuracy; 3) multitemporal object-based CD, which combines multitemporal images for unified segmentation and calculates similarity based on the features of segmented objects to achieve CD; and 4) hybrid-based CD, which integrates various spatial information using different approaches, surpassing the limitations of a single method.

However, as the quality of RS images continues to improve, the traditional methods often lack sufficient accuracy and reconstructive capabilities to meet practical application needs, particularly for land-use CD tasks where there is a multitude of diverse objects.

B. DL-Based CD Methods

These years, with the development of DL, a large number of RS image processing methods based on DL have been proposed [26] and applied in several fields such as images, sounds, and texts. The DL-based methods are also widely used in the fields of RS images' processing [27] and show excellent performance [28], such as image registration [29], road extraction [30], [31], image-to-map translation [6], [32], and image segmentation [33], and show excellent performance. Because of the powerful feature extraction and representation capabilities, CD methods based on DL demonstrate superior performance compared with the traditional methods. Among these methods, the most popular architecture for RS CD tasks is the UNet based on convolutional neural networks (CNNs) [34]. The original UNet [35] is a symmetric encoder–decoder structure that extracts feature in the encoder part using CNN and downsampling operations, and it reconstructs the image and outputs the final CD results in the decoder part using CNN and upsampling operations [36]. The skip connections between the encoder and the decoder of UNet allow for the fusion of deep semantic information and shallow spatial information, thereby improving the accuracy of CD results.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:10:49 UTC from IEEE Xplore. Restrictions apply.

UNet used in the RS CD task has many extensions [37], [38], [39]. Peng et al. [40] proposed Nested UNet for CD task by adding more intermediate convolutional blocks and skip connections between the encoder and the decoder of UNet, reducing the semantic gap of the features extracted by the encoder. According to the characteristic that the inputs of RS CD networks are bitemporal images, Daudt et al. [41] proposed a Siamese UNet by combining the Siamese networks with UNet, making the encoder of UNet become a dual-channel structure with shared parameters that extract features from the input two images, respectively. The extracted features are then fused together before fed into the decoder. This improvement enhances the network's ability to compare the differences between bitemporal images. To achieve CD of satellite images and UAV images, Shao et al. [42] modified the two branches of the encoder part of the Siamese UNet according to the size of satellite images and UAV images. Fang et al. [43] proposed a Siamese Nested UNet model called SNUNet, which integrates Siamese UNet with Nested UNet, significantly increasing the accuracy of CD results.

Using attention mechanism in CD models to improve the performance has also become a research hotspot. Zhang et al. [44] incorporated the channel attention mechanism and spatial attention mechanism in the decoder of UNet for CD tasks. The channel attention mechanism calculates the contributions of each channel of the features extracted by convolutional blocks and weights each channel according to their contributions. The spatial attention mechanism focuses on the contribution of features in the spatial dimension. These attention mechanisms make the network more effective in using information during the upsampling process. Chen and Shi [45] proposed a spatial–temporal attention-based method for RS image CD to capture the spatial–temporal correlations between individual pixels at different locations and times. The spatial–temporal attention mechanism builds upon the self-attention mechanism [46] by computing query, key, and value vectors from input features using convolutional layers. The weight assigned to value vectors is determined by an affinity function of the query and key vectors, and an output vector is obtained by the weighted sum of the value vectors. This approach fully uses the spatial and temporal dependencies between pixels to obtain illumination-invariant and misregistration-robust features. Graph attention mechanism (GAT) [47] has also gained popularity due to its powerful ability to extract and fuse long-range features. Sun et al. [21] added the graph attention mechanism before the decoder of UNet. They used the features extracted by the encoder as nodes to build a graph, representing the dependencies between each node through attention calculations. The final features are obtained by fusing the nodes.

The single-temporal supervised CD method proposed by Zheng et al. [22] recently is a novel supervised method used for building CD. The traditional supervised RS CD methods require bitemporal images and their CD labels, while the single-temporal supervised CD method can train CD models with single-temporal images and their building extraction labels. Pseudo bitemporal images are created by randomly pairing images from building extraction datasets, and the CD labels are obtained by performing XOR operation on the building extraction labels of the pseudo bitemporal images. The training datasets, composed of pseudo bi-temporal images and their labels, are used to train the single-temporal supervised CD model. The excellent test performance in the RS CD datasets proves it is effective to train CD model with single-temporal images.

C. Semi-Supervised CD Methods

In the field of DL, obtaining a large amount of data is the guarantee to train effective neural networks. In the era of big data, collecting data can be easy but producing labels for them is usually hard and time-consuming, especially pixelwise labels. SSL is one of the solutions to this limitation. It aims to use a small amount of labeled data and a large amount of unlabeled data to train models, which can save the labor and time spent on labeling to a large extent [10].

SSL has been increasingly used in the field of RS image processing. The practical application of DL-based CD methods is limited due to the difficulty in identifying and labeling changes between RS images. The SSL-based CD methods will be helpful in alleviating this problem and popularizing the DL-based CD methods. Peng et al. [20] proposed a semi-supervised CD method based on GAN. Using a generator to output the CD predictions of labeled and unlabeled RS bitemporal images, and designing two discriminators to strengthen the consistency of predictions between labeled and unlabeled images, the performance of CD can be improved with the use of unlabeled images. Zhao and Peng [17] proposed a Siamese variational autoencoder (VAE) which combines the unsupervised feature learning with supervised network fine-tuning to achieve CD in SAR images. Sun et al. [21] proposed a semi-supervised CD method based on consistency regularization and pseudolabel. First, the unlabeled bitemporal images were fed with weak augmentation into the model to obtain pseudolabels. Then, the same images were processed with strong augmentation and fed into the model to obtain predictions. By increasing the consistency of pseudolabels and predictions, the method improved the performance of the CD model.

III. Proposed Method

In this section, the architecture of the proposed semi-supervised CD method is illustrated first. Then, different components of the PITA generator with error-prone data enhancement fine-tuning strategy are presented in detail. Finally, the details of the CD networks are introduced.

A. Architecture of SemiBuildingChange

The architecture of our proposed SemiBuildingChange is illustrated in Fig. 1. It consists of two parts, the supervised part and the unsupervised part.

In the supervised part, first, N single-temporal images are randomly selected from the bitemporal RS dataset that is waiting for CD and marked as X_{st}, where st stands for single-temporal. After manually annotating the building extraction labels Y_{be} for X_{st}, X_{st} and Y_{be} are fed into the random overlay module (ROM) to generate pseudo bitemporal RS images

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:10:49 UTC from IEEE Xplore. Restrictions apply.

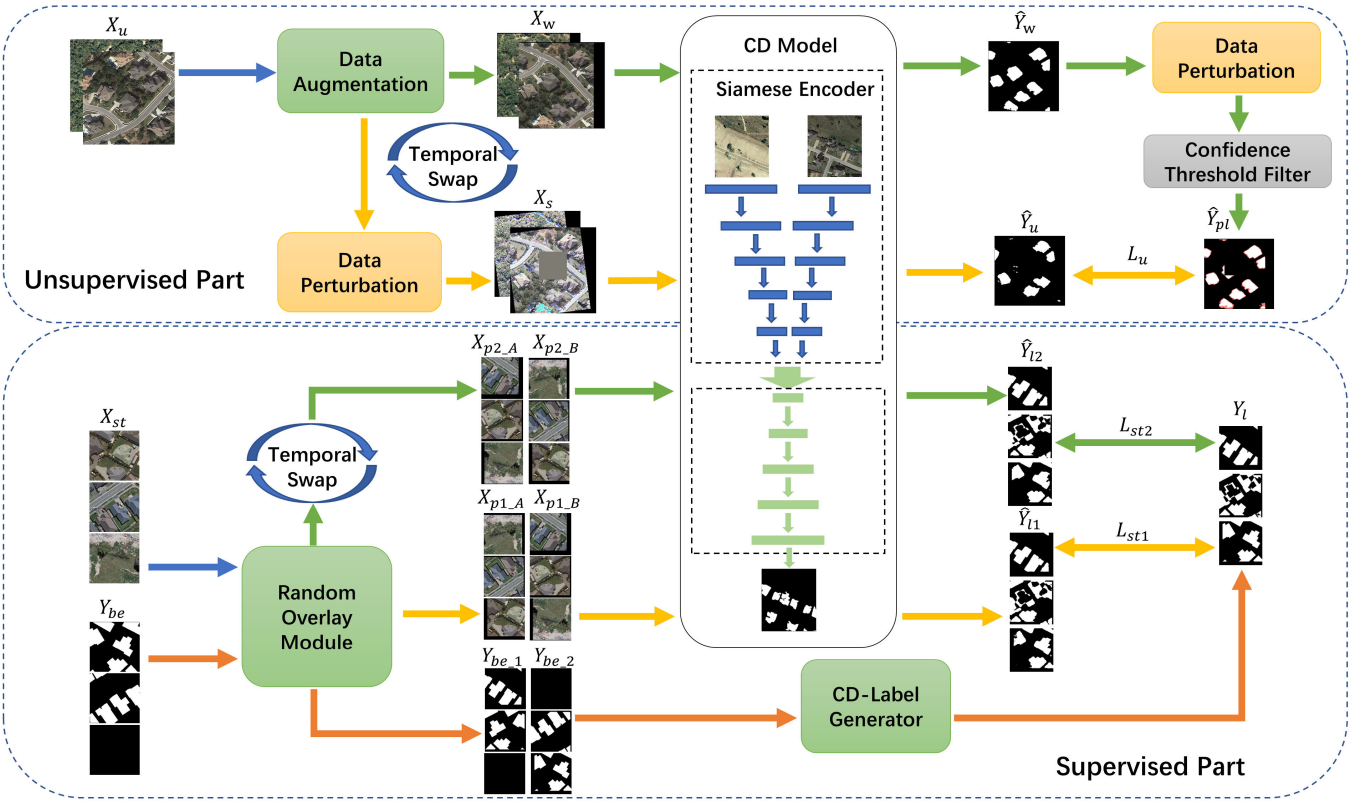


Fig. 1. Flowchart of the method SemiBuildingChange.

X_{p1_A} and X_{p1_B} and the corresponding labels $Y_{\text{be_1}}$ and $Y_{\text{be_2}}$, where be stands for building extraction, and $p1_A$ and $p1_B$ stand for pseudo bitemporal images. Then, the pseudo bitemporal images are fed into the temporal swap (Tswap) module to obtain their temporal symmetrical images X_{p2_A} and X_{p2_B}. At the same time, the building extraction labels $Y_{\text{be_1}}$ and $Y_{\text{be_2}}$ will be fed into the CD label generator to produce the final CD labels Y_l. Next, the pseudo bitemporal images X_{p1_A} and X_{p1_B} will be fed into the CD model to obtain CD results $\hat{Y}_{l1}$, and their temporal symmetrical images X_{p2_A} and X_{p2_B} will also be fed into the model to obtain CD results $\hat{Y}_{l2}$. Finally, $\hat{Y}_{l1}$ and $\hat{Y}_{l2}$ perform the loss function calculation with Y_l, respectively, to obtain the losses L_{st1} and L_{st2}. And the supervised loss L_{st} is the summarize of L_{st1} and L_{st2} as follows:

$$L_{\text{st}} = L_{\text{st1}} + L_{\text{st2}}. \tag{1}$$

In the unsupervised part, the technology of augmentation and consistency regularization mentioned in [21] is used, and the time swap module is added to further improve the performance of semi-supervised method. First, the unlabeled bitemporal RS images X_u are processed by data augmentation to produce X_w, and X_w are the data perturbed to produce X_s. To obtain CD prediction results Y_w and Y_u, X_w and X_s are fed into the CD model, respectively. Then, to generate pseudolabels $\hat{Y}_{\text{pl}}$ that correspond identically to $\hat{Y}_u$ in terms of pixel positions, and ensure the CD confidence of the pixels of $\hat{Y}_{\text{pl}}$, Y_w is processed by data perturbation and confidence threshold filter.

The loss L_{u1} is finally obtained as the loss function calculation results of $\hat{Y}_u$ and $\hat{Y}_{\text{pl}}$. X_{w_swap} is the temporal symmetrical images of X_w and obtained by the Tswap module. The process using X_{w_swap} to produce $\hat{Y}_{\text{pl_swap}}$ and $\hat{Y}_{u_\text{swap}}$ is the same as the process using X_w to produce $\hat{Y}_{\text{pl}}$ and $\hat{Y}_u$. And these two processes are conducted simultaneity. The loss L_{u2} is obtained by subjecting $\hat{Y}_{\text{pl_swap}}$ and $\hat{Y}_{u_\text{swap}}$ to a loss function calculation. And the supervised loss is the summarize of L_{u1} and L_{u2}, which can calculated as follows:

$$L_u = L_{u1} + L_{u2}. \tag{2}$$

The final loss L_{CD} is calculated from L_{st} and L_u and is used to update the parameters of the CD model by the optimizer. Finally, a highly accurate CD model is obtained and used in the inference process.

The details of the architecture are introduced below.

1) Data Augmentation and Perturbation: Data augmentation aims to increase the diversity of the images and facilitate the generalization of the model. In the SemiBuildingChange, random flipping and shifting are used as data augmentation techniques. Data perturbation [48] aims to distort the images greatly, which can increase or decrease the color gap between objects and also distort the shape of objects, causing significant changes to the original features of the images. The consistency regularization used in SSL [49] requires the model to make consistent change predictions for the RS images before and after strong perturbation, i.e., the model is required to ignore such changes brought by the perturbation and focus on the real semantic changes of interest. The data perturbation methods

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:10:49 UTC from IEEE Xplore. Restrictions apply.

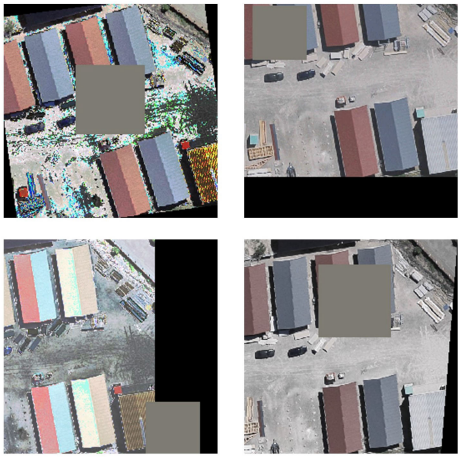


Fig. 2. Multiple data perturbation methods are randomly applied to RS images.

used in SemiBuildingChange refer to paper [21]. The effect is shown in Fig. 2.

2) Confidence Threshold Filter: In the SemiBuildingChange, the supervised part and unsupervised part are carried out simultaneously, which means the quality of the pseudolabel produced by X_w may be poor in the early stages of training. To address this issue, the confidence threshold filter, as referenced from the paper [21], is used in the unsupervised part. This filter sets a threshold τ to filter out pseudolabels with low confidence, only retaining those with high confidence. For example, if the pixel values of CD prediction results range from 0 to 1, and τ is set to 0.9, pixels with probability less than 0.9 in the pseudolabel $\hat{Y}_w$ are not included in loss function calculation. This helps prevent the negative effect of poor quality pseudolabels on the model training process.

B. PITA Generator With Error-Prone Data Enhancement Fine-Tuning Strategy

The PITA generator with the error-prone data enhancement fine-tuning strategy is the key component of SemiBuildingChange. It can synthesize a large number of pseudo bitemporal RS images using a small number of single-temporal images and select error-prone data for model fine-tuning. This component, referred to as PITA, consists of four parts: ROM, Tswap module, CD label generator, and error-prone data enhancement fine-tuning strategy, as shown in Fig. 3. The ROM and CD label generator aim to generate a large number of pseudo bitemporal images and their CD labels using single-temporal images and their building extraction labels. The Tswap module applies a temporal symmetry constraint to the model to improve the CD results. The error-prone data enhancement fine-tuning strategy records and uses error-prone data in each iteration to enhance the learning effect.

1) Random Overlay Module: ROM is the key to use a small number of single-temporal images to synthesize a large amount of pseudo bitemporal images. Through random replacement sampling and data augmentation, ROM allows for the overlaying of different positions within the same or different images to compare semantic differences. This means that the semantic differences between any positions of any images can be exploited, resulting in a much larger number of pseudo bitemporal RS images for the CD model than the input single-temporal images alone.

Suppose that the amount of input single-temporal images X_{st} is N. The ROM first performs N times random sampling on X_{st} and its building extraction label Y_{be} to obtain $X_{\text{RS_A}}$ and the corresponding label $Y_{\text{RS_A}}$. Then a random data augmentation operation is performed on $X_{\text{RS_A}}$ and $Y_{\text{RS_A}}$ to obtain X_{p1_A} and $Y_{\text{be_1}}$. At the same time, X_{p1_B} and $Y_{\text{be_2}}$ are obtained after performing another N times random sampling and random data augmentation operations on X_{st} and Y_{be}. Finally, X_{p1_A} and X_{p1_B} are paired together as pseudo bitemporal images, and $Y_{\text{be_1}}$ and $Y_{\text{be_2}}$ are paired together to further produce pseudo CD labels. In this process, the role of random sampling and random data augmentation is to enable the same or different positions in the input single-temporal RS images X_{st} to be randomly overlaid, allowing for a comparison of their semantic differences.

Using this process, the labeled data volume can grow exponentially. First, random pairing without data augmentation and allowing images to pair with themselves generate $C_N^2 + N = \frac{N(N+1)}{2}$ types of pseudo bitemporal RS images. Then, since two single-temporal RS images undergo different random data augmentation processes, the pixel regions where these two images are paired are completely random. Assuming there are M types of random data augmentation, there will be M types of pseudo bitemporal images that can be produced. So if random sampling and random data augmentation are applied to the input single-temporal RS images X_{st}, there are $M \cdot \frac{N(N+1)}{2}$ kinds of pseudo bitemporal images that can be produced theoretically. This makes the original dataset with only N single-temporal RS images expandable into a dataset with $M \cdot \frac{N(N+1)}{2}$ pairs of pseudo bitemporal RS images. This is not possible using the traditional method of labeled bitemporal RS images, because only labeled single-temporal RS images which can be freely paired can achieve such a large expansion of a small amount of labeled data.

It should be noted that random sampling used in the training process allows for replacement sampling, meaning duplicate images may be included in the sample. However, since data augmentation is applied randomly to each image, the likelihood of pseudo bitemporal RS images consisting of two identical images is small and negligible.

2) Tswap Module: The principle of temporal symmetry was proposed in the paper [22]. It states that although two bitemporal RS images have a temporal orientation in reality, their changes should be undirected for the CD network. For instance, if we consider two images T1 and T2, the prediction outcomes of the CD model should not be influenced by whether, in reality, the change is from image T1 to T2 or vice versa.

Based on the principle of temporal symmetry, a Tswap module is incorporated into the SemiBuildingChange method. Specifically, the proposed method first sends X_{p1_A} and X_{p1_B} into the CD network to obtain the predicted value $\hat{Y}_{l1}$. Then,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:10:49 UTC from IEEE Xplore. Restrictions apply.

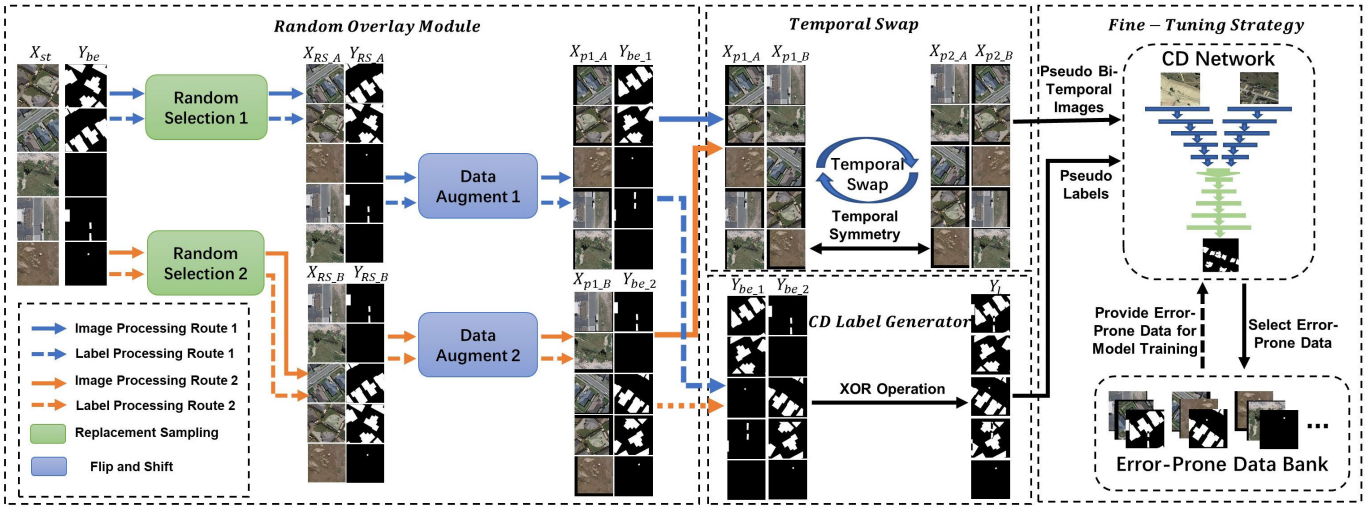


Fig. 3. Flowchart of the pseudo bitemporal generator with error-prone data enhancement fine-tuning strategy.

their temporal swap results X_{p2_A} and X_{p2_B} are fed into the CD network to obtain the predicted value $\hat{Y}_{l2}$. The principle of temporal symmetry requires the CD model to maintain consistency between $\hat{Y}_{l1}$ and $\hat{Y}_{l2}$.

It should be noted that although the two channels of the Siamese network which are used as the CD network of SemiBuildingChange share parameters when extracting features, the two feature maps are concatenated after extraction, and the concatenation is sequential. The time swap module imposes a constraint on the model that the order of concatenation should not affect the results of the model.

3) CD Label Generator: Since the labels of single-temporal RS images are building extraction labels, after generating pseudo bitemporal RS images, pseudo CD labels should also be produced using the corresponding building extraction labels.

Suppose there are two paired RS images X_A and X_B, their building extraction labels are $Y_{\text{be_1}}$ and $Y_{\text{be_2}}$, which are both binary images and the pixel value is 0 for no building and 1 for building. Obviously, the change regions are the pixel locations where buildings appear only once on X_A and X_B and can be obtained by performing an XOR operation on X_{p1_A} and X_{p1_B}. The results of the XOR operation can be used as the pseudo CD labels for X_A and X_B. Based on the above considerations, SemiBuildingChange uses the XOR operation as the CD label generator.

4) Error-Prone Data Enhancement Fine-Tuning Strategy: The PITA generator is able to produce a large number of pseudo bitemporal images by the ROM. Since the overlay of single-temporal images is completely random, there is only a small probability that the generated images will be duplicated each time, i.e., only a small number of pseudo bitemporal RS images will be trained multiple times. However, different pseudo bitemporal RS images possess different information, have different learning difficulties, and contribute differently to model training. Based on the assumption that images with a larger loss function value in each training are more difficult to train, and improving the learning effect of these images has a greater effect on the model improvement, the error-prone data enhancement fine-tuning strategy is proposed.

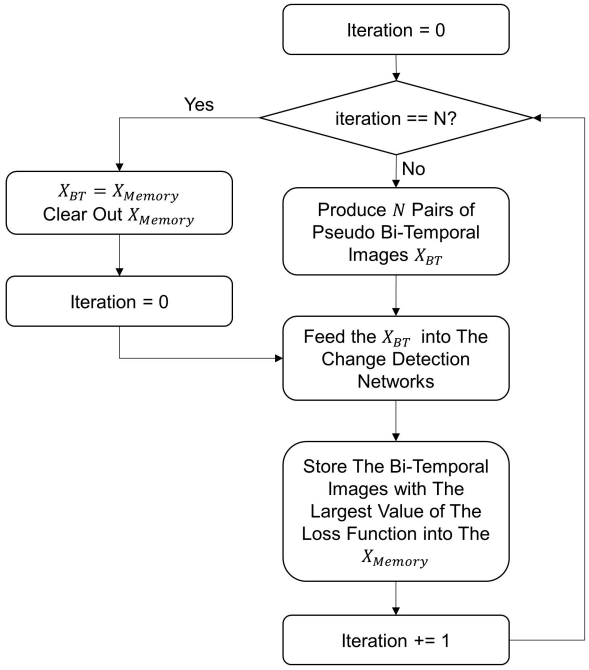


Fig. 4. Flowchart of the error-prone data enhancement fine-tuning strategy.

The specific process of the strategy is shown in Fig. 4. First, we generate N pairs of pseudo bitemporal RS images using N single-temporal RS images through the PITA generator. Then we feed these images into the CD network to get the loss function values of each pair of images, and at the same time store the images with the largest loss function value into X_{Memory}. These two steps are repeated for N rounds until there are N pairs of bitemporal RS images in X_{Memory}. Then X_{Memory} is fed into the network for another training. Note that since the input to the model is set to N pairs of RS images, every N rounds, X_{Memory} ends the collection and feeds the model

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:10:49 UTC from IEEE Xplore. Restrictions apply.

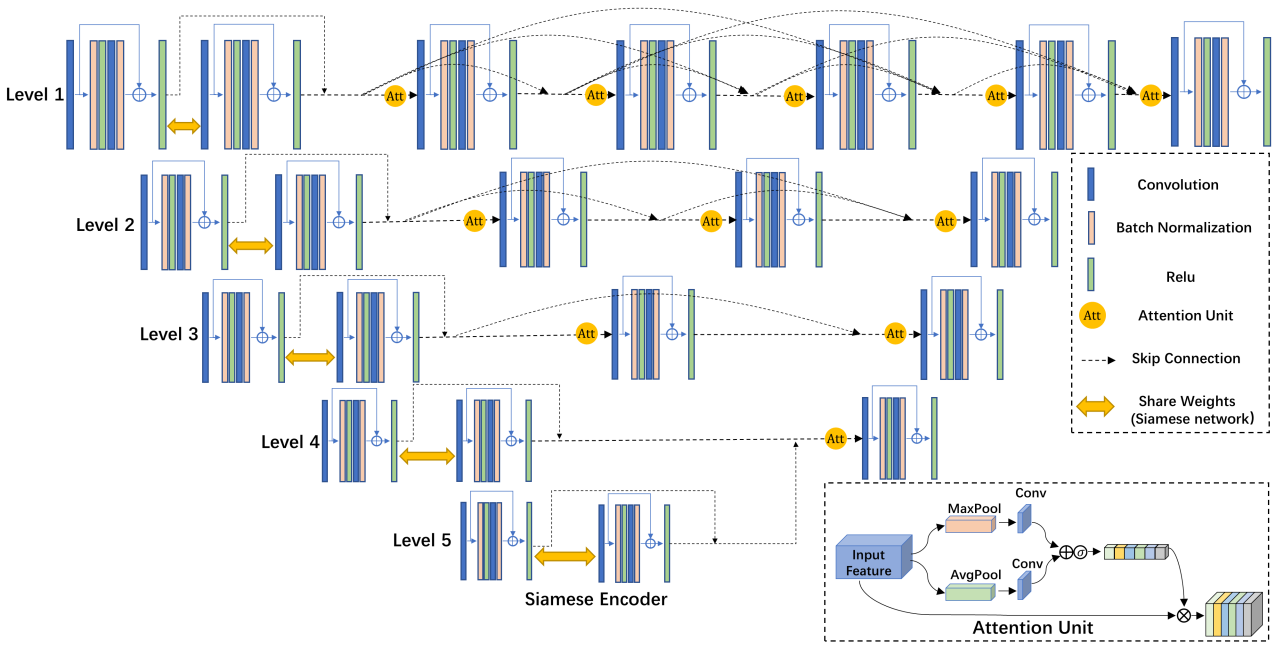


Fig. 5. Illustration of the CD network.

for training. That is, the capacity of X_{Memory} is N. Repeat the above steps until the end of the training.

C. CD Network

The CD network used in SemiBuildingChange is combined with SNUNet and dense channel attention (SNUNet-DCA), as shown in Fig. 5.

It consists of an encoder for feature extraction and a decoder for image reconstruction. The encoder is a Siamese network that has two channels, based on the characteristic that the input of the CD task are bitemporal images. Both the channels are FCN branches and share the same parameters to extract the same kind of features at the same positions of the input two RS images. Level N represents the depth of the network, where the features extracted at each level will be fed into the next level. The deeper the level, the deeper the features are extracted.

To alleviate the semantic gap between high-resolution features and fine-grained localization information, the decoder uses a strategy mentioned in [20], which involves using dense skip connections and dense channel attention between the encoder and the decoder. The input of each convolutional block in the decoder is obtained by concatenating the features extracted, decoded, and upsampled at the same level, and then feeding them into the channel attention mechanism. The channel attention mechanism assigns weights to different channels when connecting deep semantic information and shallow spatial information, thereby reducing the semantic gap between deep and shallow information.

D. Loss Function

For the CD task where the number of unchanged pixels is much larger than the number of changed pixels, we use a combination of cross-entropy loss L_{ce} and dice loss L_{dice} as the loss function.

Let the label be Y and the CD result $\hat{Y} = \hat{y}_k, k = 1, 2, \ldots, H \times W$, where H is the height of the image, W is the width of the image, and $\hat{y}_k$ is the pixel value of the kth pixel of the image. Let c be 0 or 1, representing the change or not of the kth pixel in the label. The cross-entropy loss and the dice loss are calculated as follows:

$$L_{\text{ce}} = \frac{1}{H \times W} \sum_{k=1}^{H \times W} \log\left(\frac{\exp\left(\hat{y}_{kc}\right)}{\sum_{l=0}^{1} \exp\left(\hat{y}_{kl}\right)}\right) \tag{3}$$

$$L_{\text{dice}} = 1 - \frac{2 \cdot Y \cdot \text{softmax}\left(\hat{Y}\right)}{Y + \text{softmax}\left(\hat{Y}\right)}. \tag{4}$$

So the loss function used in the SemiBuildingChange can be calculated as follows:

$$L = L_{\text{ce}} + L_{\text{dice}}. \tag{5}$$

According to formula (5), we can calculate L_{st1}, L_{st2}, L_{u1}, and L_{u2} mentioned in Section III-A and further calculate L_{st} and L_u. Let the weighting factor be λ, and the final loss L_{CD} is calculated as follows:

$$L_{\text{CD}} = L_{\text{st}} + \lambda L_u. \tag{6}$$

Obviously, λ determines the influence degree of unsupervised loss on the model, and when $\lambda = 0$, the model degenerates to a fully supervised model. We use the following pseudocode to describe our SemiBuildingChange more succinctly.

IV. Experiments

A. Datasets

To verify the effectiveness of this method, we selected two widely used RS image CD datasets: the LEVIR dataset and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:10:49 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 SemiBuildingChange Algorithm

TRAIN

Inputs : *Sigle Temporal Images* $X_{st}(Size\ is\ N)$;
Building Extraction Labels Y_{be};
Bitemporal Images Minibatch X_u;
Output : *Change Detection Model* $\mathbf{CD_{Model}}$;

while $\mathbf{CD_{Model}}$ *not converge* **do**
  $i \leftarrow 0, X_{Memory} \leftarrow None$
  for x_u *in samples of* X_u **do**
    if $i = N$ **do**
      $X_{p1_A}, X_{p1_B}, Y_l \leftarrow X_{Memory}$
      $i \leftarrow 0$
    else do
      $X_{p1_A}, X_{p1_B} \leftarrow \mathbf{ROM}(X_{st})$
      $Y_{be_1}, Y_{be_2} \leftarrow \mathbf{ROM}(Y_{be})$
      $Y_l \leftarrow \mathbf{Xor}(Y_{be_1}, Y_{be_2})$
      $i \leftarrow i + 1$
    $X_{p2_A}, X_{p2_B} \leftarrow \mathbf{Tswap}(X_{p1_A}, X_{p1_B})$
    $\hat{Y}_{l1} \leftarrow \mathbf{CD_{Model}}([X_{p1_A}, X_{p1_B}])$
    $\hat{Y}_{l2} \leftarrow \mathbf{CD_{Model}}([X_{p2_A}, X_{p2_B}])$
    $L_{st1}, L_{st2} \leftarrow \mathbf{L}(\hat{Y}_{l1}, Y_l), \mathbf{L}(\hat{Y}_{l2}, Y_l)$
    $j \leftarrow \mathbf{Argmax}(L_{st1})$
    $X_{Memory}[i] \leftarrow [X_{p1_A}[j], X_{p1_B}[j], Y_l p[j]]$
    $x_w, x_s \leftarrow \mathbf{Aug}(x_u), \mathbf{Per}(x_u)$
    $x_{w_swap} \leftarrow \mathbf{Tswap}(x_w)$
    $x_{s_swap} \leftarrow \mathbf{Tswap}(x_s)$
    $\hat{y}_w, \hat{y}_u \leftarrow \mathbf{CD}_{Model}(x_w), \mathbf{CD}_{Model}(x_s)$
    $\hat{y}_{w_swap} \leftarrow \mathbf{CD}_{Model}(x_{w_swap})$
    $\hat{y}_{u_swap} \leftarrow \mathbf{CD}_{Model}(x_{s_swap})$
    $\hat{y}_{pl} \leftarrow \mathbf{CTF}(\mathbf{Per}(\hat{y}_w))$
    $\hat{y}_{pl_swap} \leftarrow \mathbf{CTF}(\mathbf{Per}(\hat{y}_{w_swap}))$
    $L_{u1}, L_{u2} \leftarrow \mathbf{L}(\hat{y}_{pl}, \hat{y}_u), \mathbf{L}(\hat{y}_{pl_swap}, \hat{y}_{u_swap})$
    $L_{CD} = L_{u1} + L_{u2} + \lambda(L_{st1} + L_{st2})$
    Update parameters by L_{CD}
return $\mathbf{CD_{Model}}$

PREDICT

Inputs : *Bitemporal Images* X;
Output : *Change Detection Predictions* Y;

$Y \leftarrow None$
for x *in samples of* X **do**
  $y \leftarrow \mathbf{CD_{Model}}(x)$
  Y *appends* y
return *predictions* Y

the WHU dataset, both of which are the VHR datasets and contain a large number of images.

1) The LEVIR dataset (https://justchenhao.github.io/LEVIR/, accessed on 26 October 2022) is a large-scale RS BCD dataset consisting of 637 Google Earth VHR (0.5 m/pixel) images of size 1024 × 1024. The bitemporal images of the LEVIR dataset are from 20 different areas in TX, USA, and were taken from 2002 to 2018. The dataset focuses on land-use changes, primarily in buildings, and covers various types of buildings such as high-rise apartments, small garages, cottage homes, and large warehouses. The images have many pseudochanges caused by seasons, and the geometric features of buildings vary, making it a very challenging dataset. Due to the limitation of GPU memory, we segment the image pairs into 256 × 256 pixels and then divide them randomly into a training set and a test set. The training set contains 7120 images, while the test set contains 1024 images. Due to the fact that 2000 images from the LEVIR dataset are enough to train a high-precision CD model, and to save training time, we create the LEVIR2000 dataset which consists of 2000 images randomly selected from the LEVIR dataset.
2) The WHU dataset (https://study.rsgis.whu.edu.cn/pages/download/building_dataset.html, accessed on 26 October 2022) contains two RS images and their change map taken in the same area of Christchurch, New Zealand, in 2012 and 2016. Each image is 32507 × 15345 pixels in size with a resolution of 0.075 m/pixel, and the main change of interest is buildings. Due to GPU memory limitation, we segmented it by 256 × 256 pixels to obtain a training set with 2000 image pairs and a test set with 996 image pairs. The WHU dataset is a difficult dataset because of the large variation and extremely heterogeneous distribution of the changed objects.

B. Evaluation Metrics

To effectively evaluate the CD effectiveness of various methods, we used F1-Score, Kappa coefficient (Kappa), and intersection-over-union (IoU) as evaluation metrics, and they are defined as follows:

$$\mathrm{F1} = \frac{2 \times P \times R}{P + R} \tag{7}$$

$$\mathrm{Kappa} = \frac{\mathrm{OA} - \mathrm{PRE}}{1 - \mathrm{PRE}} \tag{8}$$

$$\mathrm{IoU} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN}}. \tag{9}$$

Let TP denote the number of true positives, FP denote the number of false positives, TN denote the number of true negatives, and FN denote the number of false negatives, we can calculate P, R, OA, and PRE as follows:

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{10}$$

$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}$$

$$\mathrm{PRE} = \frac{(\mathrm{TP} + \mathrm{FN}) \times (\mathrm{TP} + \mathrm{FP}) + (\mathrm{TN} + \mathrm{FP}) \times (\mathrm{TN} + \mathrm{FN})}{(\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN})^2} \tag{11}$$

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}}. \tag{12}$$

C. Experimental Setting and Baselines

The proposed method is implemented in PyTorch framework, which is installed with Intel Core i7-10700 CPU(2.9 GHz, eight cores, and 16-GB RAM) and two

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:10:49 UTC from IEEE Xplore. Restrictions apply.

NVIDIA GTX 3090 GPU with 24-GB RAM. AdamW optimizer is adopted with a learning rate 1e−3 and a weight decay 1e−2. The proposed semi-supervised method was trained for 2000 iterations both for the LEVIR dataset and the WHU dataset. The batch size is set to N for the labeled data for both the datasets, where N is the amount of labeled single-temporal images. While the batch size of the unlabeled data is set to 10 for the balance of training speed and GPU memory. According to the ablation studies' results of paper [21], the hyperparameter λ was set as 8, and the confidence threshold τ was set to 0.95. For the CD network, the number of convolution kernels of each convolution module is set to {8,16,32,64,128}, and the size of convolution kernels is set to 3×3.

To verify the effectiveness of our method, we compare the proposed method with the state-of-the-art fully supervised CD and semi-supervised CD methods, which are as follows.

1) *KPCA-MNet [13]:* KPCA-MNet is an unsupervised CD method. It extracts the high-level spatial–spectral feature maps by a deep Siamese network consisting of weight-shared kernel principal component analysis (KPCA) convolutional layers and obtains the CD results by threshold segmentation.
2) *SNUNet-ECAM [43]:* SNUNet-ECAM is a fully supervised RS CD method. It combines the Siamese network and nested UNet and effectively fuses the outputs of four nested UNets with different depths through an ensemble channel attention mechanism (ECAM).
3) *BIT-CD [50]:* BIT-CD is a fully supervised RS CD method based on the transformer. It proposes a bitemporal image transformer (BIT) to efficiently and effectively model contexts within the spatial–temporal domain.
4) *s4GAN [51]:* s4GAN is a semi-supervised semantic segmentation method. By setting the feature matching loss, it is able to use unsupervised data to minimize the difference between the predicted and true segmentation maps from semi-supervised data.
5) *semiCDNet [20]:* semiCDNet is a semi-supervised CD method using GAN. It uses two discriminators to enhance the consistency of the segmentation maps and entropy maps between labeled and unlabeled data.
6) *SemiSANet [21]:* SemiSANet is a semi-supervised CD method that uses the principle of consistency regularization to achieve high-accuracy CD results in the case of insufficient labels.
7) *ChangStar [22]:* ChangeStar is the first fully supervised method completely using single-temporal RS images with building extraction labels to train an effective CD model.

Since the input data used by the above methods are quite diverse, we have adopted the following principles to ensure fairness in our comparison. For the fully supervised method, SNUNet-ECAM, as well as the semi-supervised methods, s4GAN, semiCDNet, and semiSANet, we will use an equal number of bitemporal images as those used in the SemiBuildingChange dataset for single-temporal images. For example, if SemiBuildingChange has used 20 labeled single-temporal images, we will include 20 pairs of bitemporal images as labeled data for the above-mentioned comparison methods. It should be noted that the labels used by SemiBuildingChange are building extraction labels, while the labels used by the comparison methods are CD labels. ChangeStar used the xview2 building extraction dataset, as stated in the original paper [22]. In addition, KPCA-MNet, as an unsupervised method, will be trained without any labels.

D. Comparison Experiments

1) Prediction on LEVIR2000 Dataset: The proposed method and the baseline methods are compared in experiments on the LEVIR2000 dataset. In addition, to evaluate the time taken by each method, we recorded the duration of training for one iteration and labeled it as "TIME."

Based on the different numbers of labeled data, we calculated and summarized the quantitative results under three evaluation metrics, which are presented in Table I.

From Table I, we can see that SemiBuildingChange is significantly better than the compared unsupervised method, fully supervised methods, semi-supervised methods, and single-temporal method in the CD task on the LEVIR2000 dataset. The unsupervised method KPCA-MNet did not obtain effective BCD results on the LEVIR2000 dataset, with an F1-Score of 9.72%. This is because without labels to guide the model, KPCA-MNet can not only focus on the changes in buildings but also detect changes in other objects such as vegetation and roads on the LEVIR2000 dataset. The F1-Score of the fully supervised method SNUNet-ECAM and BIT-CD has been maintained at about 30% when the amount of labeled bitemporal RS images is less than 30 pairs. This is due to the insufficient amount of labeled data to support the training of the fully supervised model.

For the semi-supervised methods s4GAN and semiCDNet, the F1-Score of the CD results gradually improves with an increase in the number of labeled bitemporal images and significantly outperforms SNUNet-ECAM. However, their accuracy remains below 61%, which is not considered high. The semi-supervised method semiSANet has a better effect than the two methods mentioned above, with an F1-Score between 70% and 74% in experiments. However, as the amount of labeled bitemporal images increases, the performance improvement of the method is not significant. This is because the guidance effects of the labeled data on network learning are not much different when the amount of labeled data is between 10 and 30 for semiSANet. The F1-Score of the single-temporal method ChangeStar on the LEVIR2000 dataset is 72.07%. Moreover, the CD accuracy of ChangeStar is higher than the two semi-supervised methods that use GAN and close to semiSANet.

The accuracy of SemiBuildingChange is considerably better than that of SNUNet-ECAM, s4GAN, and semiCDNet. When the number of the labeled single-temporal RS images is 10, SemiBuildingChange achieves considerably higher CD accuracy than other methods, with an F1-Score of 81.67%. When the number of the labeled single-temporal RS images is 15 and 20, the F1-Score of SemiBuildingChange is 83.69% and 83.36% respectively, which is around 2% higher than

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:10:49 UTC from IEEE Xplore. Restrictions apply.

TABLE I
EXPERIMENTAL RESULTS ON THE LEVIR2000 DATASET

Method	Number of Labeled Images															
	10				15				20				30			
	F1	Kappa	IoU	TIME	F1	Kappa	IoU	TIME	F1	Kappa	IoU	TIME	F1	Kappa	IoU	TIME
KPCA-MNet	0.0972	0.7214	0.0511	440 s	0.0972	0.7214	0.0511	440 s	0.0972	0.7214	0.0511	440 s	0.0972	0.7214	0.0511	440 s
SNUNet-ECAM	0.2755	0.2570	0.1597	0.13 s	0.3390	0.3192	0.2041	0.17 s	0.3335	0.3121	0.2001	0.23 s	0.3375	0.3076	0.2030	0.37 s
BIT-CD	0.2294	0.2074	0.1295	0.42 s	0.1504	0.1351	0.0813	0.52 s	0.1911	0.1793	0.1056	0.81 s	0.3274	0.3106	0.1957	1.07 s
s4GAN	0.3295	0.2882	0.1973	99 s	0.4767	0.4479	0.3129	107 s	0.5626	0.5404	0.3914	115 s	0.5430	0.3076	0.2030	140 s
semiCDNet	0.3842	0.3502	0.2378	90 s	0.4769	0.4568	0.3131	112 s	0.6007	0.5863	0.4293	147 s	0.6049	0.5903	0.4336	178 s
semiSANet	0.7002	0.6876	0.5387	85 s	0.7132	0.7014	0.5543	90 s	0.7376	0.7272	0.5843	97 s	0.7300	0.7204	0.5749	113 s
ChangeStar	0.7207	0.7081	0.5633	25 min	0.7207	0.7081	0.5633	25 min	0.7207	0.7081	0.5633	25 min	0.7207	0.7081	0.5633	25 min
SemiBuildingChange	**0.8167**	**0.8090**	**0.6902**	**86 s**	**0.8369**	**0.8299**	**0.7195**	**94 s**	**0.8336**	**0.8265**	**0.7146**	**100 s**	**0.8443**	**0.8375**	**0.7306**	**115 s**

when the number of the labeled images is 10. When the amount of labeled single-temporal images is 30, the F1-Score of SemiBuildingChange is already 11.43% higher than semiSANet. This is because for semiSANet, when using 30 pairs of labeled images to train, it really uses 30 pairs of bitemporal images as labeled data. But for SemiBuildingChange, the 30 labeled single-temporal images are used to produce pseudo bitemporal images much more than 30 pairs using PITA generator. Although the pseudo CD labels generated by building extraction labels are not as accurate as artificial CD labels, the advantage in data volume is sufficient for the SemiBuildingChange to outperform semiSANet, especially when the data are particularly scarce.

To visualize this comparison, the CD results for each model when the amount of labeled data is 20 are shown in Fig. 6

It can be seen from Fig. 6 that SemiBuildingChange outperforms other methods in terms of both detection accuracy and object reconstruction details. The first two rows of Fig. 6 show the detection results in situations where the color characteristics of the detection object are similar to the surrounding ones. The comparison methods all have missed detection errors and false detection errors to varying degrees. Some of them judge the change buildings as surroundings, while others judge the changing surroundings as buildings. However, SemiBuildingChange is able to overcome these errors and identify these changed objects more accurately. The third and fourth rows of Fig. 6 show the CD results of each methods when there are many buildings. It can be seen that the comparison methods are inferior to SemiBuildingChange in detail reconstruction and detection accuracy. The last two rows of Fig. 6 show the CD results when the color and shape characteristics of detection objects are different from most of the objects on the LEVIR2000 dataset. There are missed detection errors in comparison methods but not in SemiBuildingChange, which also have better reconstruction results.

In terms of the training time for a single iteration, it is evident that there are significant differences among the various methods. The unsupervised method, KPCA-MNet, requires more than 7 min for training due to the utilization of the entire LEVIR2000 dataset without any labels. On the other hand, the supervised methods SNUNet-ECAM and BIT-CD have shorter training times, as they are based on a limited amount of training data in the experiments, taking less than 1 s. However, the semisupervised methods, namely, s4GAN, semiCDNet, semiSANet, and SemiBuildingChange, have considerably longer training times, ranging from 85 to 178 s. Notably, the training times of consistency-regularization-based methods semiSANet and SemiBuildingChange are relatively shorter among the semisupervised methods. Moreover, the training times of both the supervised and semisupervised methods increase as the amount of labeled data increases. In contrast, the training time for the single-temporal supervised method, ChangeStar, remains fixed and is significantly longer. This can be attributed to the fact that ChangeStar uses the entire xView2 dataset as its training dataset, which contains a larger number of images compared with the other methods.

2) Prediction on WHU Dataset: The proposed method and the baseline methods were compared in experiments conducted on the WHU dataset. Quantitative results were calculated and summarized based on different numbers of labeled data, which are presented in Table II.

From the results shown in Table II, it is evident that the SemiBuildingChange method outperforms other methods significantly on the WHU dataset. The unsupervised method, KPCA-MNet, achieves better CD results on the WHU dataset compared with the LEVIR2000 dataset, with an F1-Score of 20.87%. This is attributed to the fact that the primary changed objects in the WHU dataset are buildings. However, the effectiveness of KPCA-MNet is still inferior to that of the supervised and semi-supervised methods. The fully supervised method, SNUNet-ECAM, does not perform well on the WHU dataset due to the limited availability of labeled data, with an F1-Score consistently below 55%. As a transformer-based fully supervised method, BIT-CD achieves better results than SNUNet-ECAM due to its superior long-distance information fusing capability. Nevertheless, its F1-Score remains below 60% due to the constraint of labeled data quantity. The CD results of the GAN-based semi-supervised methods, s4GAN and semiCDNet, surpass those of SNUNet-ECAM. However, the improvement in CD accuracy is not significant with

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:10:49 UTC from IEEE Xplore. Restrictions apply.

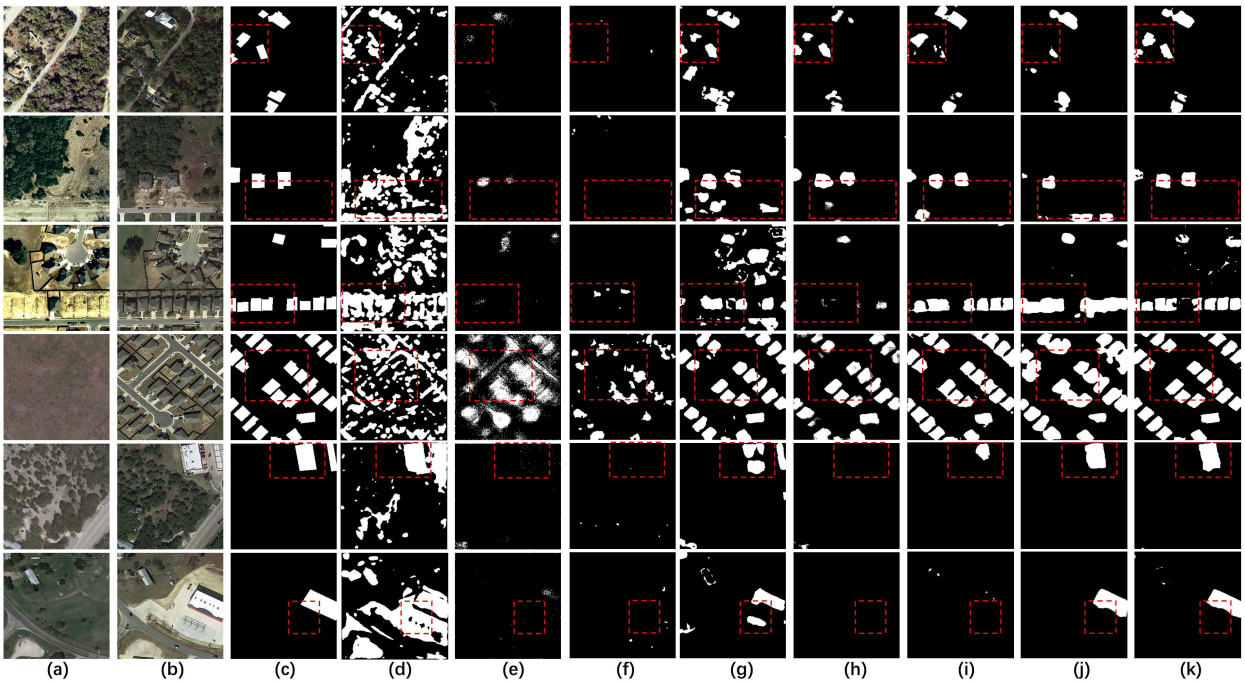


Fig. 6. Visual comparisons of CD maps by different approaches on the LEVIR2000 dataset using 20 labeled images for training. (a) Image T_1. (b) Image T_2. (c) Ground truth. (d) KPCA-MNet. (e) SNUNet-ECAM. (f) BIT-CD. (g) s4GAN. (h) semiCDNet. (i) semiSANet. (j) ChangeStar. (k) SemiBuildingChange.

TABLE II
EXPERIMENTAL RESULTS ON THE WHU DATASET

Method	Number of Labeled Images															
	10				15				20				30			
	F1	Kappa	IoU	TIME	F1	Kappa	IoU	TIME	F1	Kappa	IoU	TIME	F1	Kappa	IoU	TIME
KPCA-MNet	0.2087	0.7210	0.1165	440 s	0.2087	0.7210	0.1165	440 s	0.2087	0.7210	0.1165	440 s	0.2087	0.7210	0.1165	440 s
SNUNet-ECAM	0.3228	0.2594	0.1924	0.13 s	0.5110	0.4486	0.3432	0.17 s	0.5287	0.4674	0.3594	0.23 s	0.5444	0.4883	0.3740	0.37 s
BIT-CD	0.3849	0.2840	0.2382	0.42 s	0.5393	0.4735	0.3691	0.52 s	0.5830	0.5368	0.4114	0.81 s	0.5907	0.5451	0.4191	1.07 s
s4GAN	0.6364	0.5924	0.4667	99 s	0.5898	0.5351	0.4182	107 s	0.6197	0.5674	0.4490	115 s	0.5763	0.5182	0.4048	140 s
semiCDNet	0.6251	0.5765	0.4547	90 s	0.6265	0.5892	0.4561	112 s	0.6335	0.5853	0.4636	147 s	0.6331	0.5952	0.4631	178 s
semiSANet	0.5316	0.4870	0.3620	85 s	0.7603	0.7331	0.6132	90 s	0.7778	0.7568	0.6364	97 s	0.7812	0.7595	0.6410	113 s
ChangeStar	0.6704	0.6358	0.5042	25 min	0.6704	0.6358	0.5042	25 min	0.6704	0.6358	0.5042	25 min	0.6704	0.6358	0.5042	25 min
SemiBuildingChange	**0.8069**	**0.7855**	**0.6763**	86 s	**0.8366**	**0.8188**	**0.7191**	94 s	**0.8419**	**0.8250**	**0.7270**	100 s	**0.8579**	**0.8430**	**0.7511**	115 s

the increase in labeled bitemporal RS images, and the F1-Score remains around 63%. Concerning semiSANet, when the labeled data consist of only ten pairs of bitemporal images, the F1-Score is merely 53.16%. As the amount of labeled images increases to 15 pairs, the effectiveness of semiSANet improves considerably with an F1-Score of 76.03%. Subsequently, the F1-Score continues to improve gradually as the data quantity increases but consistently remains below 78.5%. In terms of the single-temporal supervised method, ChangeStar, it provides better CD results in situations where there is a scarcity of labeled data, with an F1-Score of 67.04%. SemiBuildingChange exhibits significantly higher CD accuracy than other semi-supervised methods. With the use of PITA generator and error-prone data enhancement fine-tuning strategy, the SemiBuildingChange method achieves excellent CD performance for each number of labeled images, with an F1-Score exceeding 80%. When only ten single-temporal images are used for labeled data, the F1-Score of SemiBuildingChange is 80.69%, which is 13.65% higher than the second best model. When using 30 labeled single-temporal images, SemiBuildingChange can achieve excellent performance with an F1-Score of 85.79%.

To visualize this comparison, the CD results of each methods when the amount of labeled data is 20 are shown in Fig. 7.

From an analysis of Fig. 7, it is evident that the WHU dataset presents greater challenges compared with the LEVIR2000 dataset. This difficulty arises from significant differences in both color characteristics and geometric features of changed objects between the two images captured in different time phases. The first two rows of Fig. 7 show the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:10:49 UTC from IEEE Xplore. Restrictions apply.

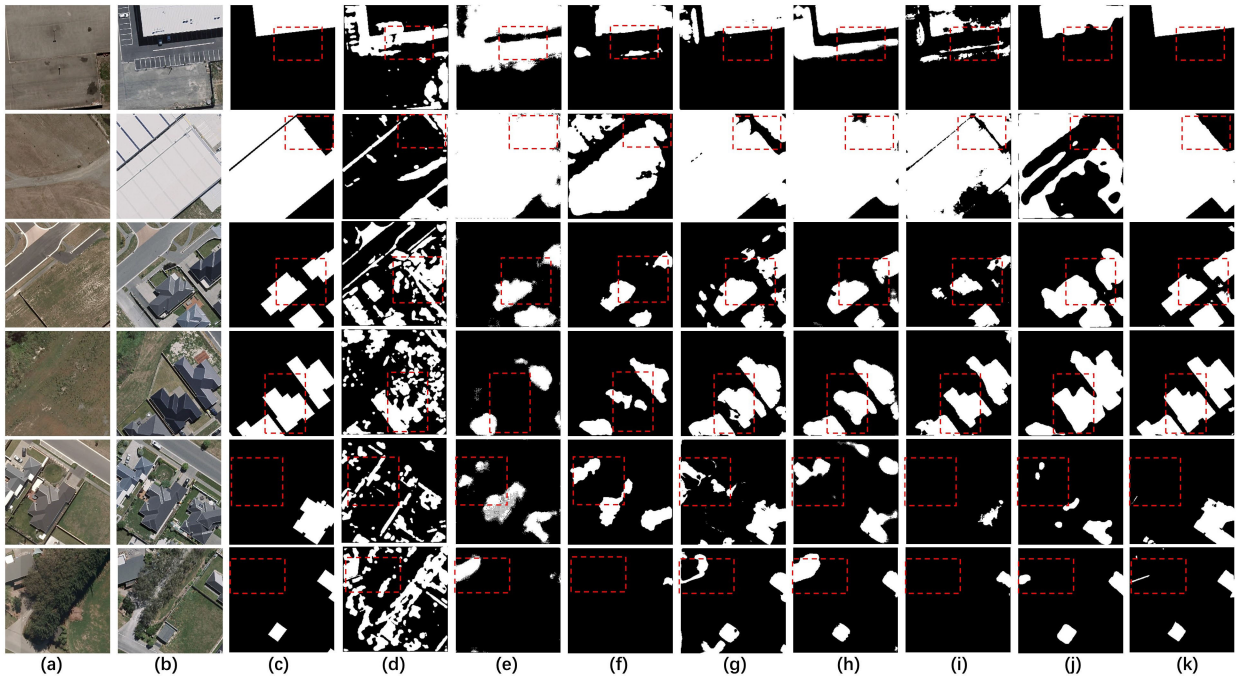


Fig. 7. Visual comparisons of CD maps by different approaches on the WHU dataset using 20 labeled images for training. (a) Image T_1. (b) Image T_2. (c) Ground truth. (d) KPCA-MNet. (e) SNUNet-ECAM. (f) BIT-CD. (g) s4GAN. (h) semiCDNet. (i) semiSANet. (j) ChangeStar. (k) SemiBuildingChange.

CD results for buildings with large areas. Due to similarities in color between the roads, land, and the changed objects, the methods tend to make false detection errors. In addition, different from the small patches which make up the majority of the dataset, the changed objects are large in size which makes it difficult for the methods to reconstruct the objects and their details. As a result, the comparison-based method exhibits varying degrees of false detections and missed detections, while SemiBuildingChange overcomes these errors. The third and fourth rows of Fig. 7 show the CD results for buildings with complex geometric features. It is apparent that SemiBuildingChange successfully reconstructs the geometric features of the changed objects, demonstrating its ability to capture intricate shapes accurately. The last two rows of Fig. 7 show the performance of each method when there is a significant color disparity between images captured in different time phases. The SNUNet-ECAM, s4GAN, semiCDNet, and ChangeStar methods mistake the surroundings with different colors as semantic changes. In contrast, SemiBuildingChange effectively avoids these errors by correctly identifying them as pseudochanges.

E. Ablation Studies

To better determine the impact of each component of the model on the effectiveness of the BCD task, ablation experiments are designed to explore the effects of the PITA generator, the error-prone data enhancement fine-tuning strategy, and the SSL component.

1) Effect of Each Component of SemiBuildingChange: To verify the effectiveness of each component of SemiBuildingChange, we conducted ablation experiments on each part of SemiBuildingChange under the condition that the amount of labeled single-temporal RS images was 20. To assess the effectiveness of the PITA generator in the semi-supervised method SemiBuildingChange, we first designated the method without the PITA generator as the original SemiBuildingChange. We then designated SemiBuildingChange with the time swap module as SemiBuildingChange + Tswap, with the ROM as SemiBuildingChange + ROM, with both the time swap module and ROM as SemiBuildingChange + Tswap + ROM, and with the PITA generator with error-prone data enhancement fine-tuning strategy as SemiBuildingChange + PITA, and conducted experiments on these methods. At the same time, to demonstrate the effectiveness of the PITA generator for the fully supervised method, we designated the supervised method without the PITA generator as SNUNet-DCA, designated the SNUNet-DCA with the time swap module as SNUNet + Tswap, with the ROM as SNUNet + ROM, with both the time swap module and ROM as SNUNet + Tswap + ROM, and with the whole PITA as SNUNet + PITA, respectively. The quantitative evaluation results are shown in Table III. The experimental results on the LEVIR2000 dataset and WHU dataset demonstrate that the PITA generator has a significant improvement effect on both the fully supervised method SNUNet-DCA and the semi-supervised method SemiBuildingChange.

In the experiments conducted on the LEVIR2000 dataset, the F1-Score of SNUNet-DCA was only 34.03% when the PITA generator was not added. However, after adding the time swap module alone, the F1-Score increased by 7.1%. Similarly, after adding the ROM alone, the F1-Score increased by 17.59%. When both the time swap module and ROM were added, the F1-Score increased to 57.68%, with an improvement of over 23%. With the error-prone data enhancement fine-tuning strategy, the F1-Score was further improved by

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:10:49 UTC from IEEE Xplore. Restrictions apply.

TABLE III
EXPERIMENTAL RESULTS OF PITA GENERATOR

Method	Semi	ROM	Tswap	Strategy	LEVIR2000 Dataset			WHU Dataset		
					F1	Kappa	IoU	F1	Kappa	IoU
SNUNet-DCA	×	×	×	×	0.3403	0.3044	0.2050	0.5370	0.4770	0.3671
SNUNet+Tswap	×	×	✓	×	0.4113	0.3810	0.2589	0.5651	0.5036	0.3938
SNUNet+ROM	×	✓	×	×	0.5162	0.4933	0.3499	0.5884	0.5316	0.4168
SNUNet+Tswap+ROM	×	✓	✓	×	**0.5768**	**0.5573**	**0.4053**	**0.6648**	**0.6237**	**0.4979**
SNUNet+PITA	×	✓	✓	✓	**0.5848**	**0.5647**	**0.4132**	**0.6861**	**0.6476**	**0.5222**
SemiBuildingChange	✓	×	×	×	0.7313	0.7199	0.5764	0.5890	0.5528	0.4175
SemiBuildingChange+Tswap	✓	×	✓	×	0.7794	0.7701	0.6386	0.6096	0.5734	0.4384
SemiBuildingChange+ROM	✓	✓	×	×	0.8054	0.7974	0.6743	0.7253	0.6904	0.5690
SemiBuildingChange+Tswap+ROM	✓	✓	✓	×	**0.8138**	**0.8055**	**0.6860**	**0.8317**	**0.8131**	**0.7119**
SemiBuildingChange+PITA	✓	✓	✓	✓	**0.8336**	**0.8265**	**0.7146**	**0.8419**	**0.8250**	**0.7270**

0.8%. In the case of the semi-supervised method, the F1-Score of SemiBuildingChange was 73.13% when the PITA generator was not added. After adding the time swap module alone, the F1-Score increased by 4.81%. On the other hand, adding the ROM alone led to an increase of 7.41% in the F1-Score. By adding both the time swap module and ROM, the F1-Score saw an improvement of 8.25%, reaching 81.38%. Finally, with the error-prone data enhancement fine-tuning strategy, the F1-Score was further improved by 1.98%.

In the experiments on the WHU dataset, the F1-Score of SNUNet-DCA is only 53.17% without the PITA generator. The F1-Score is increased by 2.81% with the addition of the time swap module and 5.14% with the ROM. When both the ROM and time swap module are added, the F1-Score increases to 66.48%, which is more than a 12% improvement. Finally, when the strategy is added, the F1-Score is further improved by 2.13%. For SemiBuildingChange, the F1-Score is 58.90% without the PITA generator. With the addition of the time swap module alone, the F1-Score is increased by 2.06%. With the ROM alone, the F1-Score is increased by 13.63%. After adding both the ROM and time swap module, the performance of SemiBuildingChange has a significant improvement, with an F1-Score of 81.38%, which is more than a 24% increase. Furthermore, the addition of the strategy improves the F1-Score by 1.98%.

It can be seen that for both the SNUNet-DCA and SemiBuildingChange models, the addition of the ROM brings much better improvement than the addition of the time swap module to the CD results, in both the datasets. This is due to the fact that the ROM is capable of generating a larger number of pseudo bitemporal images and their CD labels using a small amount of labeled single-temporal images. Thus, it greatly increases the number of labeled data and prevents the model from overfitting.

The PITA generator also plays a crucial role in enhancing the guidance of data for training the CD model, particularly on the WHU dataset. The RS images in the WHU dataset have messy object distributions and complex geometric and color characteristics, making the original SemiBuildingChange less effective than the SNUNet-DCA with a PITA generator. This is because without the PITA generator, there are insufficient labeled data to guide the training of SemiBuildingChange effectively. Even if a large amount of unlabeled bitemporal images are added, the performance improvement of the model is minimal.

2) Effect of Error-Prone Data Enhancement Fine-Tuning Strategy: To verify the effectiveness of the error-prone data enhancement fine-tuning strategy, we conducted experiments using the SemiBuildingChange method with and without the strategy at each data volume for both the datasets. It should be noted that the error-prone data enhancement fine-tuning strategy is abbreviated as strategy in the next.

The experimental results are shown in Table IV. In the experiments of the LEVIR2000 dataset, it can be observed that using the error-prone data enhancement fine-tuning strategy effectively improves the CD results of SemiBuildingChange. The strategy increases the F1-Score by more than 6% when using ten single-temporal images as labeled data, from 75.53% to 81.67%. Similarly, when using 15 single-temporal images, the addition of the strategy increases the F1-Score from 79.88% to 83.69%. For 20 single-temporal images, the strategy improves the F1-Score by 1.98%, Kappa by 2.10%, and IoU by 2.86%. Finally, for 30 single-temporal images, the strategy improves the F1-Score, Kappa, and IoU by 1.14%, 1.18%, and 1.70%, respectively.

In the experiments of the WHU dataset, it can be seen that the use of the error-prone data enhancement fine-tuning strategy significantly increases the CD results of SemiBuildingChange on the WHU dataset. When only using ten single-temporal images, the strategy is able to improve the F1-Score from 68.82% to 80.69%, Kappa from 65.10% to 78.55%, and IoU from 52.46% to 67.63%, all by more than 10%. When only using ten single-temporal images, the strategy is able to improve the F1-Score from 68.82% to 80.69%, the Kappa from 65.10% to 78.55%, and the IoU from 52.46% to 67.63%, all by more than 10%. These results demonstrate the potential of the error-prone data enhancement fine-tuning strategy to effectively improve the CD results of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:10:49 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
EXPERIMENTAL RESULTS OF ERROR-PRONE DATA ENHANCEMENT FINE-TUNING STRATEGY

Dataset	Method	Number of Labeled Images											
		10			15			20			30		
		F1	Kappa	IoU	F1	Kappa	IoU	F1	Kappa	IoU	F1	Kappa	IoU
LEVIR2000	SemiBuildingChange without strategy	0.7553	0.7457	0.6068	0.7988	0.7900	0.6651	0.8138	0.8055	0.6860	0.8329	0.8257	0.7136
	SemiBuildingChange	0.8167	0.8090	0.6902	0.8369	0.8299	0.7195	0.8336	0.8265	0.7146	0.8443	0.8375	0.7306
WHU	SemiBuildingChange without strategy	0.6882	0.6510	0.5246	0.7620	0.7340	0.6155	0.8317	0.8131	0.7119	0.8393	0.8216	0.7230
	SemiBuildingChange	0.8069	0.7855	0.6763	0.8366	0.8188	0.7191	0.8419	0.8250	0.7270	0.8579	0.8430	0.7511

SemiBuildingChange, especially when there are few images available. When using 15 single-temporal images, the strategy improves the F1-Score by 7.46%, 8.48%, and 10.36%. Similarly, when using 20 single-temporal images, the strategy improves the F1-Score by 1.02%, 1.19%, and 1.51%. Finally, when using 30 single-temporal images, the strategy can improve the F1-Score from 83.93% to 85.79%, from 82.16% to 84.30%, and from 72.30% to 75.11%.

3) Effect of Semi-Supervised Method: To demonstrate the effectiveness of the semi-supervised method, we conducted experiments using both the fully supervised method SNUNet-DCA and the semi-supervised method SemiBuildingChange at each labeled data volume. It should be noted that SNUNet-DCA is the CD network used in SemiBuildingChange. SemiBuildingChange uses both labeled and unlabeled images while SNUNet-DCA only uses labeled images.

The experimental results are shown in Table V.

In the experiments of the LEVIR2000 dataset, when using ten single-temporal images as labeled data, the semi-supervised method can increase the F1-Score of SNUNet-DCA from 33.67% to 81.67%, Kappa from 29.45% to 80.90%, and IoU from 20.24% to 69.02%, all by more than 48%. When using 15 single-temporal images as labeled data, the semi-supervised method can increase the F1-Score of SNUNet-DCA from 48.78% to 83.69%, Kappa from 45.88% to 82.99%, and IoU from 32.25% to 71.95%, all by more than 34%. When there are 20 single-temporal images as labeled data, the semi-supervised method can improve F1-Score by 25.68%, Kappa by 26.92%, and IoU by 30.93%. When there are 30 single-temporal images used as labeled data, the semi-supervised method can improve the F1-Score, Kappa, and IoU of SNUNet-DCA by 25.25%, 26.75%, and 31.03%, respectively.

The experimental results demonstrate the effectiveness of our semi-supervised method. By adding a large amount of unlabeled bitemporal images, SemiBuildingChange achieves a CD performance that significantly exceeds SNUNet-DCA, which only uses labeled data. Furthermore, the magnitude of the performance improvement becomes more significant when the amount of labeled data is smaller. This is because our approach uses techniques such as data perturbation and consistent regularization to improve the robustness and generalization of the model by effectively exploiting the large amount of available unlabeled data. Specifically, by forcing the model to make consistent judgments on unlabeled bitemporal RS images before and after distortions, the model achieves performance that far exceeds fully supervised models, even when only a small amount of labeled data are available.

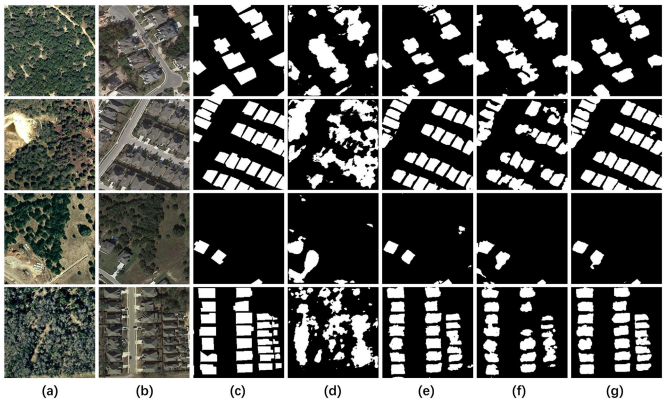


Fig. 8. Visual comparison of SNUNet-DCA and SemiBuildingChange for CD maps on the LEVIR2000 dataset. (a) Image T_1. (b) Image T_2. (c) Ground truth. (d) SNUNet-DCA with ten labeled images. (e) SemiBuildingChange with ten labeled images. (f) SNUNet-DCA with 20 labeled images. (g) SemiBuildingChange with 20 labeled images.

To see this improvement more visually, we show the results of SNUNet-DCA and SemiBuildingChange with labeled data amounts of 10 and 20 in Fig. 8.

As can be seen from Fig. 8, using only ten labeled single-temporal images, the CD results of SNUNet-DCA are very coarse. They can roughly identify the changed areas but can barely reconstruct the changed objects. However, after adding 2000 pairs of bitemporal RS images and training the model with the semi-supervised method, SemiBuildingChange not only successfully detected the changed objects but also reconstructed the changed objects in detail. When using 20 labeled single-temporal RS images, the CD results of SNUNet-DCA are better than only when using ten labeled images. This improvement is due to the use of a pseudo-data generator which increases the number of pseudo bitemporal RS images in a square order with the increase in single-temporal images However, the results of SNUNet-DCA still have some problems such as blurred contours, false detection errors, and missed detection errors. Using the SemiBuildingChange method, these problems are solved, and the details of the reconstructed changed objects become clear.

In the experiments with the WHU dataset, the CD results of SNUNet-DCA can achieve an F1-Score of 62%–66% although

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:10:49 UTC from IEEE Xplore. Restrictions apply.

TABLE V
EXPERIMENTAL RESULTS OF SEMI-SUPERVISED METHOD

Dataset	Method	Number of Labeled Images											
		10			15			20			30		
		F1	Kappa	IoU	F1	Kappa	IoU	F1	Kappa	IoU	F1	Kappa	IoU
LEVIR2000	SNUNet-DCA	0.3367	0.2945	0.2024	0.4878	0.4588	0.3225	0.5768	0.5573	0.4053	0.5918	0.5700	0.4203
	SemiBuildingChange	0.8167	0.8090	0.6902	0.8369	0.8299	0.7195	0.8336	0.8265	0.7146	0.8443	0.8375	0.7306
WHU	SNUNet-DCA	0.6616	0.6254	0.4943	0.6268	0.5811	0.4564	0.6648	0.6237	0.4979	0.6583	0.6191	0.4907
	SemiBuildingChange	0.8069	0.7855	0.6763	0.8366	0.8188	0.7191	0.8419	0.8250	0.7270	0.8579	0.8430	0.7511

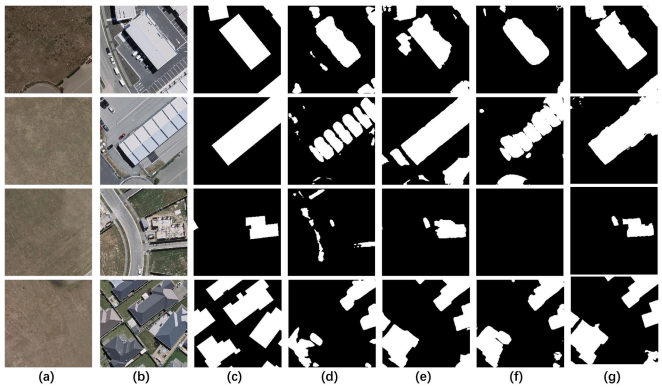


Fig. 9. Visual comparison of the SNUNet-DCA and SemiBuildingChange methods for CD maps on the WHU dataset. (a) Image T_1. (b) Image T_2. (c) Ground truth. (d) SNUNet-DCA with ten labeled images. (e) SemiBuildingChange with ten labeled images. (f) SNUNet-DCA with 20 labeled images. (g) SemiBuildingChange with 20 labeled images.

only a small amount of labeled single-temporal RS images are used to train the model. This is due to the fact that as a single-temporal supervised method using PITA generator, SNUNet-DCA is able to generate a large number of pseudo bitemporal images with a small amount of single-temporal images. Thus, SNUNet-DCA can achieve a decent accuracy even when data are scarce. After using a large number of unlabeled bitemporal RS images, the SemiBuildingChange method improves the F1-Score of CD results by 14.53%, 20.98%, 17.71%, and 19.96% with 10, 15, 20, and 30 labeled single-temporal RS images, respectively, which are significant improvements.

To see this improvement more intuitively, we show the fully supervised and semi-supervised results in Fig. 9.

As can be seen from Fig. 9, though the labeled single-temporal RS images used as training dataset are scarce, the SNUNet-DCA method with PITA generator is still able to detect the changed objects and reconstruct the contours and details of changed objects, while there are some missed detection errors. After adding a large number of unlabeled bitemporal RS images, semi-supervised method SemiBuildingChange can solve these missed detection errors successfully and perform better in reconstructing change object in detail.

V. DISCUSSION

A. Effect of the Overlay Size

In the ROM of SemiBuildingChange, two single-temporal images and their building extraction labels are overlaid after randomly shifting and flipping to produce pseudo bitemporal images and pseudo CD labels. In this process, the overlay size is determined by the extent of randomly shifting. To test the effect of overlay size, we conducted experiments on both the LEVIR2000 dataset and the WHU dataset.

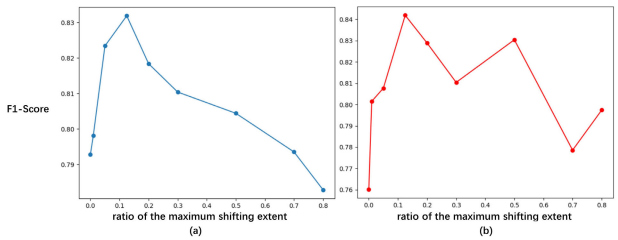


Fig. 10. Effect of the overlay size. (a) LEVIR2000 dataset. (b) WHU dataset.

We set the ratio of the maximum shifting extent as $\{0, 0.01, 0.05, 0.125, 0.2, 0.3, 0.5, 0.7, 0.8\}$. It should be noted that when the ratio is r and the size of image is N, the shifting size is randomly selected from 0 to $r \times N$. Setting the extent of shifting as $[0, r \times N]$ rather than $r \times N$ can enhance the diversity of overlay.

The experimental results are presented in Fig. 10. It is evident that when the ratio is 0, meaning no shifting operation is performed in the process, the F1-Score of SemiBuildingChange is below 80% on the LEVIR2000 dataset and below 76% on the WHU dataset. As the ratio increases, the F1-Score shows a rapid improvement, peaking at a ratio of 0.125. At this point, the F1-Score reaches 83.36% on the LEVIR2000 dataset and 84.19% on the WHU dataset. This improvement can be attributed to the increasing shifting extent ratio, which allows the PITA generator to generate a wider range of distinct bitemporal data. As the ratio continues to increase, the F1-Score on the LEVIR2000 dataset gradually decreases. Similarly, on the WHU dataset, the F1-Score shows a downward trend with jitter. This can be attributed to the fact that when the shifting extent becomes too large, the overlay size becomes too small to generate meaningful pseudo bitemporal labels. Consequently, the generated wrong pseudolabels can mislead the CD network, leading to a decrease in performance.

B. Effect of Model-Generated Labels

In the previous training, the building labels of single-temporal RS images we used were accurately labeled by hand with high quality. Naturally, it is easy to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:10:49 UTC from IEEE Xplore. Restrictions apply.

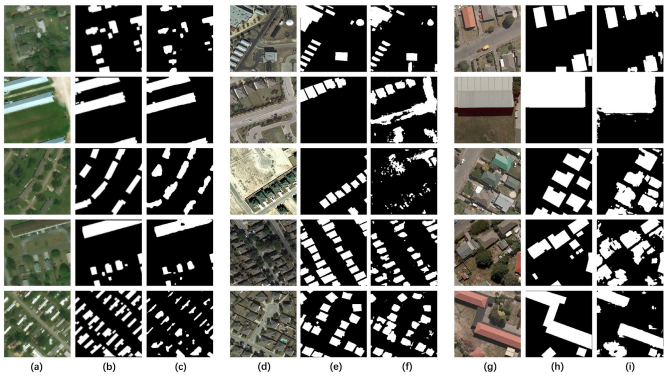


Fig. 11. Building extraction results on the xView2 dataset, LEVIR dataset, and WHU dataset. (a) Images on the xView2 dataset. (b) Ground truth. (c) Model prediction. (d) Images on the LEVIR dataset. (e) Ground truth. (f) Model prediction. (g) Images on the WHU dataset. (h) Ground truth. (i) Model prediction.

think that if these hand-made labels could be replaced by model-generated labels, then our method would be able to be an unsupervised method. We conduct a simple experiment here to examine the effect of using model-generated labels on the performance of CD.

The building extraction model used is SSTNet proposed by Chen et al. [52]. First, we trained SSTNet on the xView2 building extraction dataset. Its F1-Score on the xView2 test set can reach 87%. Then we use the trained Nested UNet model to extract buildings from 20 single-temporal RS images selected from each of the LEVIR dataset and WHU dataset, and some of the results are shown in Fig. 11.

As can be seen from Fig. 11, due to the differences in the feature distribution of different datasets, the building extraction model trained on the xView2 dataset still suffers from some false detection errors and missed detection errors on the LEVIR dataset and the WHU dataset. Obviously, the quality of the model-generated building extraction results is worse than that of the manually produced building extraction labels.

Then SemiBuildingChange is trained with the hand-made labels and the model-generated labels, and the results are recorded in Table VI. It can be seen from Table VI that the quality of labels significantly influences the CD effects of SemiBuildingChange. On the LEVIR dataset, the F1-Score of hand-made labels is 18.25% higher than the F1-Score of the model-generated labels. On the WHU dataset, the F1-Score of hand-made labels is 10.17% higher than the F1-Score of the model-generated labels. This experiment shows that the quality of the building extraction labels influences the accuracy of CD results significantly, and hand-made labels get much better CD results than the model-generated labels. Since the demand for the amount of building extraction labels is quite a few in SemiBuildingChange, the training data with man-made labels can be prepared without much effort. Moreover, we can see the F1-Score of model-generated labels still exceeds 65% on the LEVIR2000 dataset and even reaches 74.02% on the WHU dataset, which is much better than the unsupervised CD method KPCA-MNet. So as an unsupervised method, SemiBuildingChange with model-generated labels has potential for development and can be further studied.

TABLE VI

EFFECT OF HAND-MADE LABELS AND MODEL-GENERATED LABELS, WHEN THE NUMBER OF LABELS IS 20

Quality of labels	LEVIR2000 Dataset			WHU Dataset		
	F1	Kappa	IoU	F1	Kappa	IoU
hand-made	0.8336	0.8265	0.7146	0.8419	0.8250	0.7270
model-generated	0.6511	0.6359	0.4827	0.7402	0.7113	0.5876

TABLE VII

EFFECT OF DATA AUGMENTATION TECHNOLOGIES, WHEN THE NUMBER OF LABELS IS 20

Method	LEVIR2000 Dataset			WHU Dataset		
	F1	Kappa	IoU	F1	Kappa	IoU
SemiBuildingChange-noAugPer	0.4993	0.4801	0.3327	0.7420	0.7124	0.5898
SemiBuildingChange-noPer	0.5409	0.5188	0.3707	0.7610	0.7356	0.6142
SemiBuildingChange-noAug	0.7696	0.7594	0.6255	0.8322	0.8141	0.7126
SemiBuildingChange	**0.8336**	**0.8265**	**0.7146**	**0.8419**	**0.8250**	**0.7270**

C. Effect of Data Augmentation Technologies

In the proposed SemiBuildingChange method, two types of data augmentation technologies were used: normal data augmentation techniques such as shifting and flipping, and data perturbation techniques such as posterization, brightness adjustment, and shearing. The former aims to enhance the diversity of PITA, while the latter serves as a vital component of SSL based on consistency regularization. To evaluate the effectiveness of these augmentation technologies, ablation experiments were conducted on both the LEVIR2000 dataset and the WHU dataset to compare the impact before and after their inclusion. The results, shown in the Table VII, demonstrate that augmentation technologies significantly improve the precision of CD outcomes.

On the LEVIR2000 dataset, not using any augmentation technologies resulted in an F1-Score of 49.93%. By incorporating data augmentation and data perturbation techniques, the F1-Scores increased by 4.16% and 27.03%, respectively. The utilization of both the augmentation technologies further elevated the F1-Score by 33.43%. Similarly, on the WHU dataset, the F1-Score without any augmentation was 74.2%. The addition of data augmentation enhanced the F1-Score by 1.9%, while the inclusion of data perturbation increased it by 9.02%. When both the augmentation technologies were used, the F1-Score improved by 9.99%. These results confirm the efficacy of augmentation technologies in improving CD performance.

D. Effect of Single-Temporal Data Volume

To test the effect and cost of single-temporal RS images' data volume on the CD results of SemiBuildingChange, we conducted experiments on both the LEVIR2000 dataset and WHU dataset. We set the amount of single-temporal images as {5, 10, 15, 20, 25, 30}. And for both the LEVIR dataset and WHU dataset, the amount of unlabeled RS images is

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:10:49 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
EFFECT AND COST OF SINGLE-TEMPORAL DATA VOLUME

ST Data Volume	GPU	TIME	LEVIR2000 Dataset			WHU Dataset		
			F1	Kappa	IoU	F1	Kappa	IoU
5	25.395 G	84 s	0.7380	0.7092	0.5848	0.7870	0.7622	0.6489
10	27.699 G	86 s	0.8167	0.8090	0.6902	0.8069	0.7855	0.6763
15	30.008 G	94 s	0.8369	0.8299	0.7195	0.8366	0.8188	0.7191
20	32.301 G	100 s	0.8336	0.8265	0.7146	0.8419	0.8250	0.7270
25	35.150 G	106 s	0.8311	0.8239	0.7109	0.8580	0.8427	0.7513
30	37.594 G	115 s	0.8443	0.8375	0.7306	0.8599	0.8454	0.7542

always 2000 pairs. The results of the experiments are shown in Table VIII, including GPU memory usage and time consumed per training batch.

It can be seen that in both the experiments on the LEVIR2000 dataset and WHU dataset, the CD results of SemiBuildingChange exhibited gradual improvement as the amount of labeled data increased. On the LEVIR2000 dataset, with the labeled single-temporal images increased from 5 to 20, the F1-Score improved rapidly from 73.73% to 81.38%. Then the increase slows down, and finally the F1-Score reaches 83.29% when the amount of labeled data is 30. On the WHU dataset, the F1-Score improves rapidly when the labeled data volume is less than 20, increasing from 60.08% to 83.17%. When the amount of labeled data exceeds 20, the performance of SemiBuildingChange is jittery, and finally, the F1-Score reaches 83.93% at the labeled data amount of 30. It can be seen that the performance of SemiBuildingChange on two datasets has similar improvement trend. They both first increase rapidly with the number of single-temporal RS images, and then enter a slowly increasing period with slight jitter. The GPU memory occupied by this method increases as the amount of single-temporal data increases. When using five single-temporal images, this method requires 25.395G of GPU memory. When using five single-temporal images, GPU memory usage has been increased to 37.594G. The amount of GPU memory occupied limits the increase in the amount of single-temporal images.

E. Effect of Unlabeled Bitemporal Data Volume

To test the effect of unlabeled bitemporal data volume to the CD performance of SemiBuildingChange, experiments are conducted on the LEVIR dataset and WHU dataset. Previously, we used only 2000 pairs of images from the LEVIR dataset as the training set, but in this experiment we used up to 4000 images from the LEVIR dataset as the training set to test the effect of a larger amount of unsupervised data. We set the amount of unlabeled bitemporal images for the LEVIR dataset to {100, 200, 400, 600, 1000, 2000, 3000, 4000}, and to {100, 200, 400, 600, 1000, 1500, 2000} for the WHU dataset. While the amount of labeled single-temporal is always 20. The results of the experiments are shown in Fig. 12.

It can be seen from Fig. 12 that on both the LEVIR dataset and WHU dataset, the F1-Score of the CD results improves

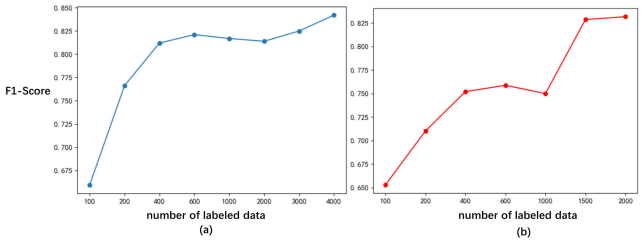


Fig. 12. Effect of unlabeled bitemporal data volume. (a) LEVIR dataset. (b) WHU dataset.

with the increase in the amount of unlabeled bitemporal RS images, although there is some jitter in the process. On the LEVIR dataset, the F1-Score increased rapidly from 65.94% to 81.19% when the amount of unlabeled images increased from 100 pairs to 400 pairs, then the F1-Score fluctuated between 81% and 82%. After using 3000 pairs of unlabeled bitemporal RS images, the F1-Score reaches 82.58%. When 4000 pairs of unlabeled images were used, the F1-Score further increased to 84.18%. It shows that on the LEVIR dataset, using more unlabeled bitemporal RS images can bring a more obvious improvement in the model performance.

On the WHU dataset, the F1-Score of SemiBuildingChange was always below 80% until the use of 1500 pairs of unlabeled RS images. The F1-Score first grew and then slowly jittered with the increase in unlabeled data. When 1500 pairs of unlabeled images were used, the F1-Score broke through to 82.86%. After using 2000 pairs of unlabeled RS images, the F1-Score further increased to 83.17%.

From the experiments on the two datasets, it can be seen that the larger the unlabeled data volume, the better the performance of the CD model obtained. Although there may be some jitter in the process, increasing the input of unlabeled bitemporal RS images can make the model break through the jitter and continue to improve. This is due to the fact that the model can learn more accurate and comprehensive distribution features from the dataset with larger amount of unlabeled data.

VI. CONCLUSION

In this article, we propose a semi-supervised method for detecting building changes in RS images called SemiBuildingChange. The method is based on a PITA generator with an error-prone data enhancement fine-tuning strategy. SemiBuildingChange is capable of using a small amount of single-temporal RS images with building extraction labels and a large amount of unlabeled bitemporal RS images to train a high-precision BCD model. Extensive experimental comparisons demonstrate that SemiBuildingChange outperforms state-of-the-art methods in cases of extreme lack of labeled data. It is worth noting that the idea of the PITA generator can be facilely extended to other object CD tasks. In future work, we plan to focus on improving this method by designing an unsupervised CD method based on the PITA generator. If building extraction labels can be produced using a high-precision building extraction model instead of manually labeling them, this method can become a high-precision unsupervised method.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:10:49 UTC from IEEE Xplore. Restrictions apply.

ACKNOWLEDGMENT

The authors would like to thank the anonymous referees for their valuable comments and helpful suggestions.

REFERENCES

[1] W. Shi, M. Zhang, R. Zhang, S. Chen, and Z. Zhan, "Change detection based on artificial intelligence: State-of-the-art and challenges," *Remote Sens.*, vol. 12, no. 10, p. 1688, May 2020.

[2] A. Madasa, I. R. Orimoloye, and O. O. Ololade, "Application of geospatial indices for mapping land cover/use change detection in a mining area," *J. Afr. Earth Sci.*, vol. 175, Mar. 2021, Art. no. 104108.

[3] Z. Lv, X. Li, W. Wang, B. Zhang, J. Hu, and S. Feng, "Government affairs service platform for smart city," *Future Gener. Comput. Syst.*, vol. 81, pp. 443–451, Apr. 2018.

[4] F. Huang, L. Chen, K. Yin, J. Huang, and L. Gui, "Object-oriented change detection and damage assessment using high-resolution remote sensing images, Tangjiao Landslide, Three Gorges Reservoir, China," *Environ. Earth Sci.*, vol. 77, no. 5, pp. 1–19, Mar. 2018.

[5] H. U. Dike, Y. Zhou, K. K. Deveerasetty, and Q. Wu, "Unsupervised learning based on artificial neural network: A review," in *Proc. IEEE Int. Conf. Cyborg Bionic Syst. (CBS)*, Oct. 2018, pp. 322–327.

[6] J. Song, J. Li, H. Chen, and J. Wu, "MapGen-GAN: A fast translator for remote sensing image to map via unsupervised adversarial learning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 2341–2357, 2021.

[7] Z.-H. Zhou, "A brief introduction to weakly supervised learning," *Nat. Sci. Rev.*, vol. 5, no. 1, pp. 44–53, Jan. 2018.

[8] S. Wu, C. Du, H. Chen, Y. Xu, N. Guo, and N. Jing, "Road extraction from very high resolution images using weakly labeled OpenStreetMap centerline," *ISPRS Int. J. Geo-Inf.*, vol. 8, no. 11, p. 478, Oct. 2019.

[9] H. Chen, S. Peng, C. Du, J. Li, and S. Wu, "SW-GAN: Road extraction from remote sensing imagery using semi-weakly supervised adversarial learning," *Remote Sens.*, vol. 14, no. 17, p. 4145, Aug. 2022.

[10] X. Yang, Z. Song, I. King, and Z. Xu, "A survey on deep semi-supervised learning," *IEEE Trans. Knowl. Data Eng.*, vol. 109, no. 2, pp. 1–20, 2022.

[11] H. Chen, Z. Li, J. Wu, W. Xiong, and C. Du, "SemiRoadExNet: A semi-supervised network for road extraction from remote sensing imagery via adversarial learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 198, pp. 169–183, Apr. 2023.

[12] J. Song, H. Chen, C. Du, and J. Li, "Semi-MapGen: Translation of remote sensing image into map via semisupervised adversarial learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4701219.

[13] C. Wu, H. Chen, B. Du, and L. Zhang, "Unsupervised change detection in multitemporal VHR images based on deep kernel PCA convolutional mapping network," *IEEE Trans. Cybern.*, vol. 52, no. 11, pp. 12084–12098, Nov. 2022.

[14] S. H. Khan, X. He, F. Porikli, M. Bennamoun, F. Sohel, and R. Togneri, "Learning deep structured network for weakly supervised change detection," 2016, *arXiv:1606.02009*.

[15] J. Dai, K. He, and J. Sun, "BoxSup: Exploiting bounding boxes to supervise convolutional networks for semantic segmentation," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1635–1643.

[16] D. Lin, J. Dai, J. Jia, K. He, and J. Sun, "ScribbleSup: Scribble-supervised convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 3159–3167.

[17] G. Zhao and Y. Peng, "Semisupervised SAR image change detection based on a Siamese variational autoencoder," *Inf. Process. Manag.*, vol. 59, no. 1, Jan. 2022, Art. no. 102726.

[18] A. Diaz-Pinto, A. Colomer, V. Naranjo, S. Morales, Y. Xu, and A. F. Frangi, "Retinal image synthesis and semi-supervised learning for glaucoma assessment," *IEEE Trans. Med. Imag.*, vol. 38, no. 9, pp. 2211–2218, Sep. 2019.

[19] F. Yang, H. Zhang, and S. Tao, "Semi-supervised classification via full-graph attention neural networks," *Neurocomputing*, vol. 476, pp. 63–74, Mar. 2022.

[20] D. Peng, L. Bruzzone, Y. Zhang, H. Guan, H. Ding, and X. Huang, "SemiCDNet: A semisupervised convolutional neural network for change detection in high resolution remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5891–5906, Jul. 2021.

[21] C. Sun, J. Wu, H. Chen, and C. Du, "SemiSANet: A semi-supervised high-resolution remote sensing image change detection model using Siamese networks with graph attention," *Remote Sens.*, vol. 14, no. 12, p. 2801, Jun. 2022.

[22] Z. Zheng, A. Ma, L. Zhang, and Y. Zhong, "Change is everywhere: Single-temporal supervised object change detection in remote sensing imagery," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 15173–15182.

[23] M. K. Ridd and J. Liu, "A comparison of four algorithms for change detection in an urban environment," *Remote Sens. Environ.*, vol. 63, no. 2, pp. 95–100, Feb. 1998.

[24] N. Falco, P. R. Marpu, and J. A. Benediktsson, "A toolbox for unsupervised change detection analysis," *Int. J. Remote Sens.*, vol. 37, no. 7, pp. 1505–1526, Apr. 2016.

[25] G. Chen, G. J. Hay, L. M. T. Carvalho, and M. A. Wulder, "Object-based change detection," *Int. J. Remote Sens.*, vol. 33, no. 14, pp. 4434–4457, 2012.

[26] S. Dong, P. Wang, and K. Abbas, "A survey on deep learning and its applications," *Comput. Sci. Rev.*, vol. 40, May 2021, Art. no. 100379.

[27] L. Ma, Y. Liu, X. Zhang, Y. Ye, G. Yin, and B. A. Johnson, "Deep learning in remote sensing applications: A meta-analysis and review," *ISPRS J. Photogramm. Remote Sens.*, vol. 152, pp. 166–177, Jun. 2019.

[28] G. Cheng, X. Xie, J. Han, L. Guo, and G.-S. Xia, "Remote sensing image scene classification meets deep learning: Challenges, methods, benchmarks, and opportunities," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 3735–3756, 2020.

[29] Y. Xu, J. Li, C. Du, and H. Chen, "NBR-Net: A nonrigid bidirectional registration network for multitemporal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5620715.

[30] Y. Xu, H. Chen, C. Du, and J. Li, "MSACon: Mining spatial attention-based contextual information for road extraction," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5604317.

[31] K. Yan, Y. Xu, and H. Chen, "Building boundary vectorization from satellite images using generative adversarial networks," in *Proc. Int. Conf. Image, Vis. Intell. Syst.* Cham, Switzerland: Springer, 2022, pp. 671–678.

[32] J. Song, J. Li, H. Chen, and J. Wu, "RSMT: A remote sensing image-to-map translation model via adversarial deep transfer learning," *Remote Sens.*, vol. 14, no. 4, p. 919, Feb. 2022.

[33] W. Zi, W. Xiong, H. Chen, J. Li, and N. Jing, "SGA-Net: Self-constructing graph attention neural network for semantic segmentation of remote sensing images," *Remote Sens.*, vol. 13, no. 21, p. 4201, Oct. 2021.

[34] T. Chen, Z. Lu, Y. Yang, Y. Zhang, B. Du, and A. Plaza, "A Siamese network based U-Net for change detection in high resolution remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 2357–2369, 2022.

[35] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.* Berlin, Germany: Springer, 2015, pp. 234–241.

[36] E. Bousias Alexakis and C. Armenakis, "Evaluation of UNet and UNet++ architectures in high resolution image change detection applications," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 43, pp. 1507–1514, Aug. 2020.

[37] M. S. Moustafa, S. A. Mohamed, S. Ahmed, and A. H. Nasr, "Hyperspectral change detection based on modification of UNet neural networks," *J. Appl. Remote Sens.*, vol. 15, no. 2, Jun. 2021, Art. no. 028505.

[38] S. Sun, L. Mu, L. Wang, and P. Liu, "L-UNet: An LSTM network for remote sensing image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[39] C. Sun, C. Du, J. Wu, and H. Chen, "SUDANet: A Siamese UNet with dense attention mechanism for remote sensing image change detection," in *Pattern Recognition and Computer Vision*. Shenzhen, China: Springer, 2022, pp. 78–88.

[40] D. Peng, Y. Zhang, and H. Guan, "End-to-end change detection for high resolution satellite images using improved UNet++," *Remote Sens.*, vol. 11, no. 11, p. 1382, Jun. 2019.

[41] R. C. Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

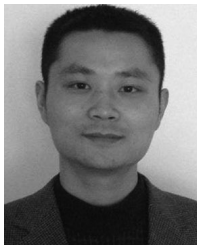
Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:10:49 UTC from IEEE Xplore. Restrictions apply.

[42] R. Shao, C. Du, H. Chen, and J. Li, "SUNet: Change detection for heterogeneous remote sensing images from satellite and UAV using a dual-channel fully convolution network," *Remote Sens.*, vol. 13, no. 18, p. 3750, Sep. 2021.
[43] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[44] C. Zhang et al., "A deeply supervised image fusion network for change detection in high resolution bi-temporal remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 183–200, Aug. 2020.
[45] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020.
[46] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–11.
[47] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Lio, and Y. Bengio, "Graph attention networks," 2017, *arXiv:1710.10903*.
[48] E. D. Cubuk, B. Zoph, J. Shlens, and Q. V. Le, "Randaugment: Practical automated data augmentation with a reduced search space," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 3008–3017.
[49] K. Sohn et al., "FixMatch: Simplifying semi-supervised learning with consistency and confidence," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 596–608.
[50] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607514.
[51] S. Mittal, M. Tatarchenko, and T. Brox, "Semi-supervised semantic segmentation with high- and low-level consistency," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 4, pp. 1369–1379, Apr. 2021.
[52] K. Chen, Z. Zou, and Z. Shi, "Building extraction from remote sensing images with sparse token transformers," *Remote Sens.*, vol. 13, no. 21, p. 4441, Nov. 2021.



Chengzhe Sun received the B.S. degree in communication engineering and the M.S. degree in information and communication engineering from the National University of Defense Technology, Changsha, China, in 2020 and 2022, respectively, where he is currently pursuing the Ph.D. degree with the College of Electronic Science and Technology.

His research interests include machine learning, remote sensing (RS) image processing, and photogrammetry.



Hao Chen received the M.S. and Ph.D. degrees in information and communication engineering from the National University of Defense Technology, Changsha, China, in 2010.

He is currently a Professor with the College of Electronic Science and Technology, National University of Defense Technology. His research interests include data mining, machine learning, and evolutionary computation.



Chun Du received the Ph.D. degree in electronic science and technology from the National University of Defense Technology, Changsha, China, in 2014.

He is an Associate Professor at the National University of Defense Technology. His research interests include machine learning, machine vision, and remote sensing (RS).



Ning Jing received the Ph.D. degree in computer science from the National University of Defense Technology, Changsha, China, in 1990.

He is currently a Professor at the National University of Defense Technology. His research interests include management and analysis of big data and spatial information system.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:10:49 UTC from IEEE Xplore. Restrictions apply.